Conjunctive gates in engineered proteins: a review of mechanisms and boundary cases

Koichiro Saka

The Protein Engineering Grammar describes proteins whose function is gated by a controlled input, using four terms: Sensor, Gate, Processor, and Actuator. Two Gate values require conjunction of inputs, which we call "conjunctive gates": AND-like, requiring multiple inputs together, and Conditional Unlock, in which a single input switches on a function that is off by default. Their molecular implementation and achieved specificity had not been reviewed across modalities.

We reviewed technologies spanning therapeutics, synthetic biology, de novo design, and biomaterials, with both Gate values well represented. Trigger chemistries range from kinase phosphorylation and endogenous small molecules to split de novo cytokines and protease logic. Several AND-like designs followed a common trajectory from partial single-input response to iteratively refined specificity, while others (e.g., AvidCAR) rely instead on low-affinity binders; both Gate values recur across modalities via diverse mechanisms.

We classified reusable trigger-recognition tactics, including sub-threshold affinities, asymmetric anchor/trigger arms, panel screening, avidity through valency, and energy-window design for competitive-displacement masks. Computational and AI-based platforms can in principle be applied to energy-window tuning and de novo generation, but remain immature at designing conformational bistability. Finally, trigger-recognition architectures fall into two binding-mode groups: one structurally constrained by trigger presentation, and one where the binding mode is itself a protein-protein interaction, offering principled extensibility beyond the small molecules and short motifs demonstrated so far.

1 Introduction

Protein therapeutics have traditionally been classified by structural format. The categories include monoclonal antibodies, bispecific antibodies, antibody-drug conjugates, Fc-fusion proteins, and cytokine-fusion proteins. This classification is useful for regulatory, manufacturing, and intellectual-property purposes. It does not, however, answer a different question: what functional logic does a molecule implement? Which conditions must be met before it functions? How does it combine several recognition events? What physical steps turn recognition into a therapeutic effect?

Two molecules with the same structural classification can implement entirely different designs along this axis. Conversely, molecules built on scaffolds unrelated to antibodies can implement the same logic as an established antibody technology. This observation motivated the Grammar [1] from which the present review draws its vocabulary.

The Protein Engineering Grammar is a complementary vocabulary for proteins that function only under certain conditions. It describes them along four axes.

- **Sensor** is the type of input a protein detects. The Grammar has six classes, and this review mainly uses three. Partner is the presence of a binding-partner molecule. Context is a condition or state of the surrounding environment. It includes enzyme activity and localization, and also physicochemical conditions such as pH, oxygen level, and light. Identity is recognition of a specific molecular identity.

- **Gate** is the logical relationship between the Sensor's input condition(s) and the expression of function.
- **Processor** is the molecule's internal mechanism that operates the Actuator once the Gate permits it.
- **Actuator** is the molecule's effect on external factors.

Gate has several classes. Two of them are the subject of this review. AND-like designs express function only when multiple input conditions are met, and each of them is independently needed. Conditional Unlock designs express function when a single input condition is met.

The Grammar's own definition splits AND-like into two sub-classes, Structural and Affinity-driven (Figure 1); Section 2 defines each in detail. In what follows, we use the term "conjunctive gates" as shorthand for AND-like and Conditional Unlock together. This is our own shorthand, not a Grammar term. Conditional Unlock requires a single condition. We use the term for what the two share: function appears only once the required input condition is met.

These two Gate classes are core mechanisms for building molecular switches. They are shared by therapeutics and synthetic biology alike. In therapeutics, part of their importance is that they widen the therapeutic window by tying function to disease-associated conditions.

Protease-dependent Conditional Unlock (masked antibodies) [2] is one example. It activates a drug only inside a tumor microenvironment, where protease activity is high. Exposure in normal tissue remains low, and potency at the tumor site is preserved.

AND-like recognition of two antigens is another example. Some normal tissues express only one of two tumor-associated antigens. A two-antigen therapy spares these tissues and kills only tumor cells that express both [3], [4]. This avoids the on-target/off-tumor toxicity that is unavoidable when a therapy targets a single antigen [5], [6].

Later sections show that the same Gate mechanism, meaning the logic of a permission condition for function, has been implemented by different companies on different scaffolds. This is consistent with the view that the mechanism is a portable design principle. The evidence reviewed here does not establish whether these implementations arose independently or by transfer of an earlier concept.

Nevertheless, how reported technologies actually implement these Gates had not been reviewed at the level of the protein architecture of individual technologies. Which modality is used? By what molecular mechanism? At what engineering cost? With what degree of specificity and activity? Existing reviews each focus on a specific modality or trigger chemistry.

- Nolan-Stevaux [6] proposes a logic-gate taxonomy (AND, OR, and contextual AND). Its scope is limited to two modalities: multispecific T-cell engagers and CAR-T cells. It does not use the formal Sensor/Gate/Processor/Actuator vocabulary. Nor does it use the Structural/Affinity-driven distinction within AND-like that this review adopts.
- Kim [7] reviews activation mechanisms of antibody platforms: masking, structural rearrangement, and ADC payload activation. It covers protease, pH, and redox triggers. However, it is limited to antibodies. It does not address de novo-designed proteins, cytokines, enzymes, or AND-like mechanisms based on multi-antigen recognition.
- Bressler [8] discusses Boolean logic in synthetic biology and biomaterials. Its focus is application to cell therapy and living materials. It does not examine the protein architecture or engineering difficulty of individual molecules.

The Grammar paper [1] is the source of this review's vocabulary, and we refer to it as the prior review. It gave an overview of molecular implementation across all Gate values. This review differs in depth. It narrows its scope to the two conjunctive gates and goes further into the protein architecture of individual technologies. Specifically, it examines the binding mode of trigger-recognition modules and structural topologies such as Cage-Latch-Key, Assembly, and avidity. It also examines the engineering tactics and the principled extensibility that follow from them.

This review therefore proceeds in four steps.

1. It organizes patterns of molecular implementation for AND-like and Conditional Unlock across modalities.

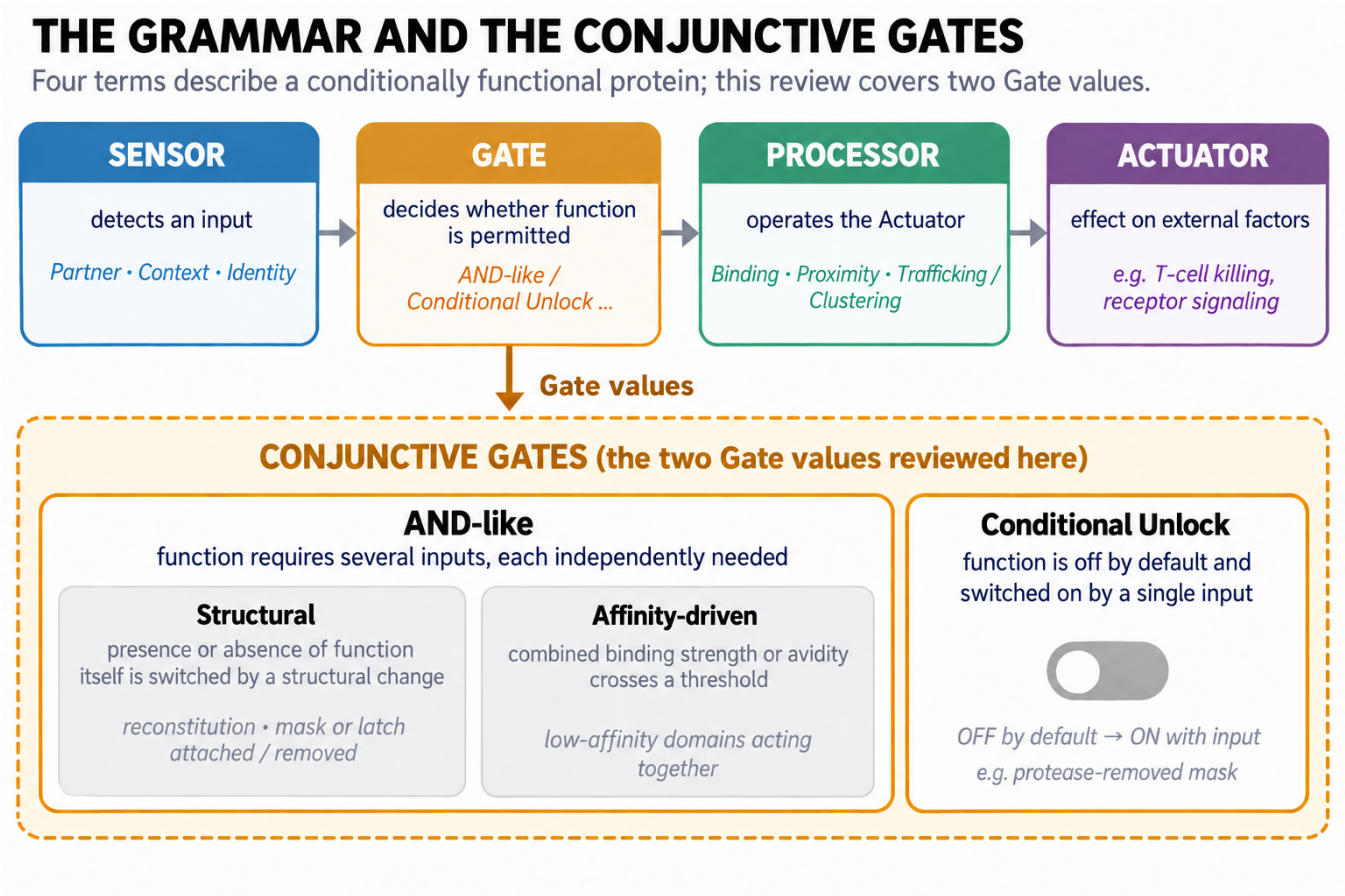


Figure 1: The Grammar's four terms, and the two Gate values reviewed here. Sensor detects an input, Gate decides whether function is permitted, Processor is the mechanism by which the molecule operates the Actuator, and Actuator is the molecule's effect on external factors. AND-like and Conditional Unlock, together called conjunctive gates, are the subject of this review.

2. It documents the degree of specificity and activity reported for individual technologies, and the iterative design process that led to it.
3. It records boundary cases next to Threshold-like and OR-like. This is a preliminary and incidental contribution to clarifying Gate definitions.
4. It classifies the protein architecture of trigger-recognition modules into reusable design tactics. It examines the current role of computational and AI-based design platforms in carrying out these tactics. It also examines the principled extensibility of trigger chemistry.

2 Overview: The Diversity of Mechanisms

We describe each technology along three axes.

(a) The modality and molecular mechanism through which it is implemented.

(b) The degree of specificity and activity achieved. We use the primary quantitative measures reported in the literature, such as fold-selectivity, leakiness, EC50 ratios, Hill coefficients, and in vivo efficacy.

(c) Where reported, the iterative process by which the design evolved from its first prototype to its current form.

Axes (b) and (c) are limited to evidence-based statements of fact. Interpretation that generalizes across several examples is left to the discussion sections that follow.

Within each Gate, we group the examples by the mechanism that implements the Gate, that is, how its condition is physically realized. Examples are

reconstitution, mask removal, competitive displacement, and avidity. This grouping is the review's own device. It is not the Grammar's Processor, which records how the molecule hands its function to the Actuator. Some of the names we use, such as Assembly, Release, Conformational change, and Clustering, are also names of Processor classes in the Grammar. Here they name how the Gate condition is implemented. They are not Processor values. In the Grammar, mask removal and reconstitution are recorded as the implementation of a Gate, not as a Processor. In the table, we call this the implementation mechanism.

Table 1 lists representative examples with the performance reported for each.

AND-like

AND-like requires several inputs, each independently needed to release the OFF state of the function. Function need not be lost completely when one input is missing. AND-like has two mechanistically distinct sub-classes.

The **Structural** sub-class works through a change in the structure of the molecule. These changes include reconstitution of a complex, and the attachment or removal of a mask or latch. They switch the physical presence or absence of function itself. How much function remains when one input is missing is not part of the definition. For example, XPAT retains intermediate activity when only one of its two masks is removed.

The **Affinity-driven** sub-class works through changes in binding strength or avidity, from the additive or multiplicative combination of multiple Sensors. Binding strength is intrinsically continuous, so when one input is missing, function often diminishes but does not disappear entirely.

The sub-class is assigned by mechanism, not by measured behavior. The values in Table 1 do not separate the two sub-classes. For example, MP0632 (Structural) shows weak or absent killing of single-antigen cells, and AMG 305 (Affinity-driven) shows greatly attenuated activity. These cannot be told apart by behavior. In LINK CAR, a Structural design, the first prototype was leaky. Among the registered technologies, some have no sub-class assigned, or one assigned by inference.

The next two subsections give examples of each sub-class. They are organized by the mechanism each example uses. Figure 2 illustrates four representative mechanisms.

Structural sub-class

Examples of the Structural sub-class fall into two groups by mechanism. One group uses physical reconstitution of a complex (Assembly). The other uses competitive attachment or removal of a mask or latch.

Reconstitution of antibodies and T-cell engagers. Hemibody [3] is a soluble molecule made of two half-antibodies. Each half fuses one variable region of an anti-CD3 antibody, either the heavy chain (VH) or the light chain (VL), to a separate tumor-antigen-specific scFv. Neither half can bind CD3 alone. When both halves co-localize on the same cell surface, VH and VL associate and reconstitute a functional anti-CD3 Fv. This engages a T cell.

The trigger is the simultaneous expression of two distinct tumor antigens (Identity Sensors). The CD3-binding site is not a single scFv. It is an antibody variable domain split across VH and VL.

A CD38 × SLAMF7 version targets multiple myeloma [9]. It shows that this design avoids fratricide, even when T cells themselves express the target antigen. Fratricide is the killing that occurs in cell therapy when cells expressing the same target antigen attack one another.

Reconstitution of CAR signaling. LINK CAR [10] replaces CD3ζ with TCR-proximal signaling molecules, LAT and SLP-76. Each is carried on a separate CAR. A GADS-mediated signaling scaffold forms only when both antigens are bound.

The first prototype was leaky and responded partially to a single antigen. True AND behavior was acquired only after iterative adjustment. The GADS-binding site was deleted, and the CD28 transmembrane domain was mutated. LINK CAR follows the same mechanistic principle as Hemibody: two Identity inputs cooperate through reconstitution (Assembly). It acts, however, on the receptor side, not in a soluble antibody.

Table 1: Representative technologies, the inputs they respond to, and the performance reported in the primary literature. The implementation mechanism is this review's own grouping and does not give the Grammar's Processor value. Values are those reported by the original authors and are not directly comparable across technologies.

Technology	Gate	Inputs	Implementation mechanism	Reported performance
Hemibody [3], [9]	AND-like (Structural)	Two tumor antigens	Assembly	Avoids fratricide, even when T cells express the target antigen
LINK CAR [10]	AND-like (Structural)	Two antigens	Assembly	First prototype was leaky; true AND behavior after GADS-site deletion and CD28 transmembrane mutation
Split Neoleukin-2/15 [11]	AND-like (Structural)	Two tumor antigens	Assembly	Complete responses in an in vivo dual-antigen tumor model
Split-Flp [12]	AND-like (Structural)	Two small molecules (gibberellin, abscisic acid)	Assembly	125-fold increase in activity (at one of two split-site pairs tested)
ProNotch [13]	AND-like (Structural)	Two proteases	Assembly	80-fold induction
Co-LOCKR [14]	AND-like (Structural)	Two antigens (Cage and Key)	Competitive displacement	Colocalization-dependent AND-like behavior
MP0632 [15]	AND-like (Structural)	Two antigens (MSLN, EpCAM)	Competitive displacement	Weak or absent killing of single-antigen and healthy primary cells; dose-dependent killing of double-positive cells
XPAT [16]	AND-like (Structural)	Protease (two masks)	Release	About 4-log protection with both masks; one mask leaves intermediate activity
AvidCAR [17]	AND-like (Affinity-driven)	Two antigens	Assembly (avidity)	Monomer-inactive domain (Kd 877 nM); dependence lost at higher affinity (Kd 13 and 125 nM)
JMB2403 [18]	AND-like (Affinity-driven)	PD-1 and IL-2R$\beta\gamma$	Assembly (avidity)	29-fold (PD-1 arm inactivated) and 42-fold (receptor saturation) differences in activity
BVX001 [19]	AND-like (Affinity-driven)	CD33 and CD7	Assembly (avidity)	Activity falls with competing Fabs; no toxicity toward healthy progenitor cells
ISB 1442 [20], [21]	AND-like (Affinity-driven)	CD38 and CD47	Assembly (avidity)	No binding to CD38-knockout cells or erythrocytes
AMG 305 [22]	AND-like (Affinity-driven)	CDH3 and MSLN	Assembly (avidity)	Activity greatly attenuated against single-antigen cells
Ly6E $\times$ B7-H4 engager [23]	AND-like (Affinity-driven)	Ly6E and B7-H4	Assembly (avidity)	Graded: the second antigen adds a modest enhancement
pro-CPG2 [24]	Conditional Unlock	Protease	Release	Graded: about 80% inhibition; 98% with disulfide bonds, but less restoration by protease
HAP18 [25]	Conditional Unlock	Hypoxia (a Context)	Release	Same mechanism as protease-based systems, with a non-protease Context Sensor
Switch-Ig [26]	Conditional Unlock	Extracellular ATP	Conformational change	Antigen binding restored only in the presence of ATP
Hinge module [27]	Conditional Unlock	Effector peptide	Assembly	Hill coefficients of 2.7-3.9; 10 of 17 soluble strained designs switched (26 attempted)
FAP-4-1BBL [28], [29]	Conditional Unlock	FAP (localization marker)	Clustering	Activity below the detection limit without the anchor
GEN1046 [30]	Conditional Unlock	Localization marker	Clustering	Activity below the detection limit without the anchor
ABBV-428 [31]	Conditional Unlock	Localization marker	Clustering	Graded and leaky: weaker but not abolished without the anchor

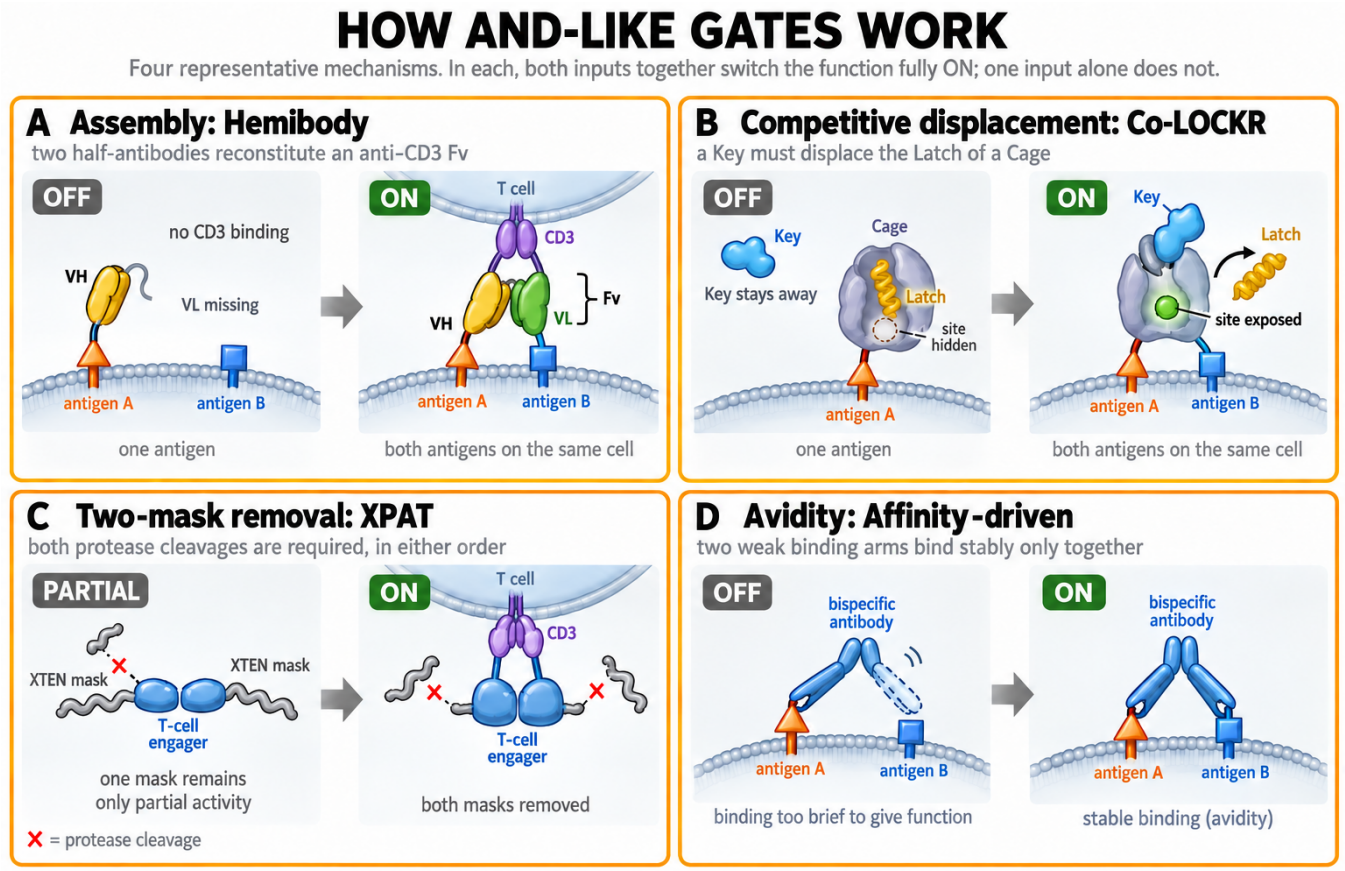


Figure 2: How AND-like gates work. Panels A-C show the Structural sub-class. They are reconstitution of an anti-CD3 Fv (Hemibody), competitive displacement of a Latch by a Key (Co-LOCKR), and removal of two independent masks (XPAT). With one XPAT mask left, activity is partial. Panel D shows the Affinity-driven sub-class, in which two weak binding arms bind stably only when both antigens are present. In each panel, both inputs together switch the function fully ON, and one input alone does not. Schematics are simplified and not to scale.

Splitting a de novo-designed cytokine. Split Neoleukin-2/15 [11] divides a de novo-designed IL-2/IL-15-mimetic cytokine into two fragments. Each fragment is fused to a protein that binds a different tumor antigen. Neither fragment activates the receptor alone, even at high concentration. When both fragments co-localize on a tumor cell that expresses both antigens, a functional receptor complex is reconstituted. The design achieves complete responses in an in vivo model of dual-antigen tumors. It extends the same principle to a new kind of building block: a de novo-designed cytokine instead of an antibody.

Protein logic in a single chain or molecule. Several examples implement logic within one chain or one molecule.

- SPOC [32] uses a pair of orthogonal proteases derived from potyviruses, together with designed coiled-coils. An autoinhibitory peptide normally blocks reconstitution of a split effector, such as split luciferase. When the trigger protease cleaves a linker, a displacer peptide replaces the autoinhibitory peptide. Reconstitution can then proceed.
- A split-Flp recombinase logic gate [12] divides Flp recombinase into three fragments. It adds two chemically inducible dimerization domains, one gibberellin-responsive and one abscisic-acid-responsive. Reconstitution occurs only when both small molecules are present. Activity then rises 125-fold. This AND behavior was seen at one of the two split-site pairs tested.
- ProNotch [13] self-circularizes the negative regulatory region of the Notch1 receptor using Spy-

Tag/SpyCatcher [33]. Two protease-cleavable linkers within a single chain hold it in this conformation. It activates (80-fold induction) only when both proteases have cleaved.

- CIPHR [34] uses de novo-designed heterodimeric proteins. A split transcription factor reconstitutes only when a specific combination of input proteins is present. This is a two-input AND gate.

These four examples share the same principle: reconstitution, or Assembly.

In the combined light-and-drug category, TamPA-Cre [35] is a split Cre recombinase. It requires two things: tamoxifen-driven nuclear translocation of an ER-localization domain, and blue-light-driven dimerization.

Competitive release of a mask or latch. Competitive release of a mask or latch, combined with co-localization, is a third principle. It is distinct from both Assembly and Release.

Co-LOCKR [14] uses a de novo-designed Cage protein. Its built-in Latch covers a functional site, which acts as a self-mask. An external Key protein exposes the functional site, but only by competitively displacing the Latch. The Cage and the Key each carry an independent antigen-binding domain. The system is designed so that competitive displacement proceeds efficiently only when both co-localize on the same cell surface. This yields colocalization-dependent AND-like behavior. The same platform also reports NOT-like constructs, in which a decoy domain competitively sequesters the Key. NOT-like is outside this review's Gate scope.

MP0632 is a fourth variant of this competitive mechanism. It is a Switch-DARPin T-cell engager from Molecular Partners [15]. It is a single multi-domain DARPin fusion (anti-MSLN, anti-EpCAM, a masked anti-CD3 domain, anti-CD2, and Fc). A built-in "Switch" mask blocks the anti-CD3 domain.

It differs from Co-LOCKR in that the Cage and the Key are not separate molecules. The anti-MSLN and anti-EpCAM domains are located in the same molecule. Only when both bind separate cell-surface antigens does an intramolecular structural change release the CD3 mask. It also differs from XPAT and TwoGATE, discussed below. Here mask removal is a non-covalent structural change. It is not protease cleavage.

Direct head-to-head comparisons in vitro and in vivo show weak or absent killing of cells with a single antigen and of healthy primary cells. Cells with both antigens are killed in a dose-dependent manner. The CD2 co-stimulatory domain is an incidental potentiator. It is not part of the Gate mechanism itself.

Removal of two independent masks. XPAT [16] is an AND-like example based on removing two independent masks. It flanks a HER2/EGFR × CD3 T-cell engager with independent XTEN masks [36] at the N- and C-termini. Both protease-cleavage events are required, in either order. This forms an order-independent AND gate. A molecule with only one mask removed retains intermediate cytotoxicity and T-cell activation [16], so the AND is graded. The sub-class follows from the mechanism, the removal of masks.

Revitope's TwoGATE [37] requires two things. One is reconstitution of a split CD3 paratope from two half-molecules, each binding a different antigen. The other is protease cleavage.

Other platforms adopt the same principle of removing two independent barriers. PRO-XTEN (Vir/Amunix) [38], [39] uses XTEN masks, as XPAT does. TRACTr (Janux) and POWERbody (Adagene) use peptide masks instead. The EGFR-targeted TRACTr molecule JANX008 carries two independent protease-cleavable peptide masks, one on the EGFR arm and one on the CD3 arm [40]. The HER2-targeted POWERbody molecule ADG138 also masks both the HER2 arm and the CD3 arm [41]. Whether the two masks of ADG138 share the same protease trigger is unresolved. This does not change the number of inputs, because each mask needs its own cleavage at its own site. The single-mask forms of these platforms are Conditional Unlock (Section 2.2).

Affinity-driven sub-class

Examples of the Affinity-driven sub-class come in diverse molecular formats. They share one mechanism: avidity, which arises when several binding domains bind a target at once.

The unifying design principle is as follows. Binding through a single domain does not persist long enough

to produce function. Bivalent binding (avidity) does. Bivalent binding occurs only when two target antigens co-express on the same cell. The two antigens are the triggers, and both are Identity Sensors.

Bispecific antibodies and ADCs.

- BVX001 [19] is a CD33 × CD7 bispecific antibody-drug conjugate. It has direct head-to-head evidence from a Fab-homodimer competition assay. Activity falls when unbound anti-CD33 and anti-CD7 Fabs are added as competitors. It shows no toxicity toward healthy progenitor cells.
- A P-cadherin × CDH17 antibody-drug conjugate [42] is still undergoing iterative refinement. The parental bispecific antibody responds partially even to single-antigen tumors. The authors themselves report minor non-specific binding in vivo, even for the final candidate.
- ISB 1442 [20], [21] is a CD38-biparatopic/CD47-bispecific antibody in a 2+1 format. It has direct head-to-head evidence of non-binding to CD38-knockout cells and to erythrocytes.
- Evobody [43] is a FOLR1 × BTN3A bispecific antibody. Its BTN3A arm is deliberately weakened in affinity. As a result, Vγ9Vδ2 T-cell activation through BTN3A is driven only by avidity on FOLR1-positive cells.

Trispecific T-cell engagers. These molecules carry one anti-CD3 domain and two tumor-antigen domains. AMG 305 [22] has bivalent domains against CDH3 and MSLN. It kills cells that express both antigens, and its activity against cells that express only one is greatly attenuated. TriMab [44] shows the same behavior with ROR1 and EGFR. Its ROR1 arm anchors the molecule, and its EGFR arm is tuned in affinity. The Ly6E × B7-H4 engager [23] is a graded case. Killing follows the higher-affinity arm, and the second antigen adds only a modest enhancement.

CARs. AvidCAR [17] combines a monomeric CAR scaffold with a low-affinity binding domain. It selectively kills double-positive cells. Dependence on antigen density is built directly into the mechanism.

Cis-activating cytokine and immunomodulatory designs. JMB2403 [18] is a PD-1 × IL-2Rβγ trispecific antibody. A high-affinity PD-1 arm (the anchor) tethers an intrinsically weak IL-2Rβγ agonist arm (the trigger) near activated T/NK cells. Inactivating the PD-1 arm with a dummy reveals a 29-fold difference in activity. Receptor-saturation experiments reveal a 42-fold difference. Selectivity is still based on a continuous variable: the PD-1 expression level.

Conditional Unlock

We organized AND-like by mechanism, that is, by how multiple conditions are physically integrated. We organize Conditional Unlock along the same axis: which mechanism turns the satisfaction of a single condition into a change in function. The Sensor type alone does not determine the physical mechanism. We therefore note the Sensor type for each entry. Figure 3 illustrates the four mechanisms.

Release (mask/prodomain removal by covalent-bond cleavage)

Release operates by physically covering an antigen-binding site or a cytokine-receptor-binding site with a peptide mask. A protease-cleavable linker holds the mask in place [45], [46]. The trigger is the activity of a tumor-associated protease, which is a Context Sensor. When it cleaves the linker, the mask is released, and binding or activity is restored.

This is the central design theme of the masked antibody/T-cell-engager/cytokine platforms reviewed here. Examples include Probody [2], [47], masked cytokines [48], XPAT, SAFEbody/POWERbody (ADG152, ADG206, ADG138) [41], [49], [50], [51], [52], COBRA [53], [54], [55], TwoGATE, and TRACTr (JANX007, JANX008). Designs with two independent masks (XPAT, ADG138, JANX008) work through the same Release mechanism, but they are AND-like (Section 2.1).

This mechanistic class also has examples outside therapeutics.

- pro-CPG2 [24] is a computationally designed protease-activatable proenzyme. Its prodomain covers the enzyme's active site. The designed prodomains inhibit the enzyme by about 80% (reduction in kcat/KM). Adding disulfide bonds raises inhibition to 98%, but lowers the restoration of activity by protease. This is a graded

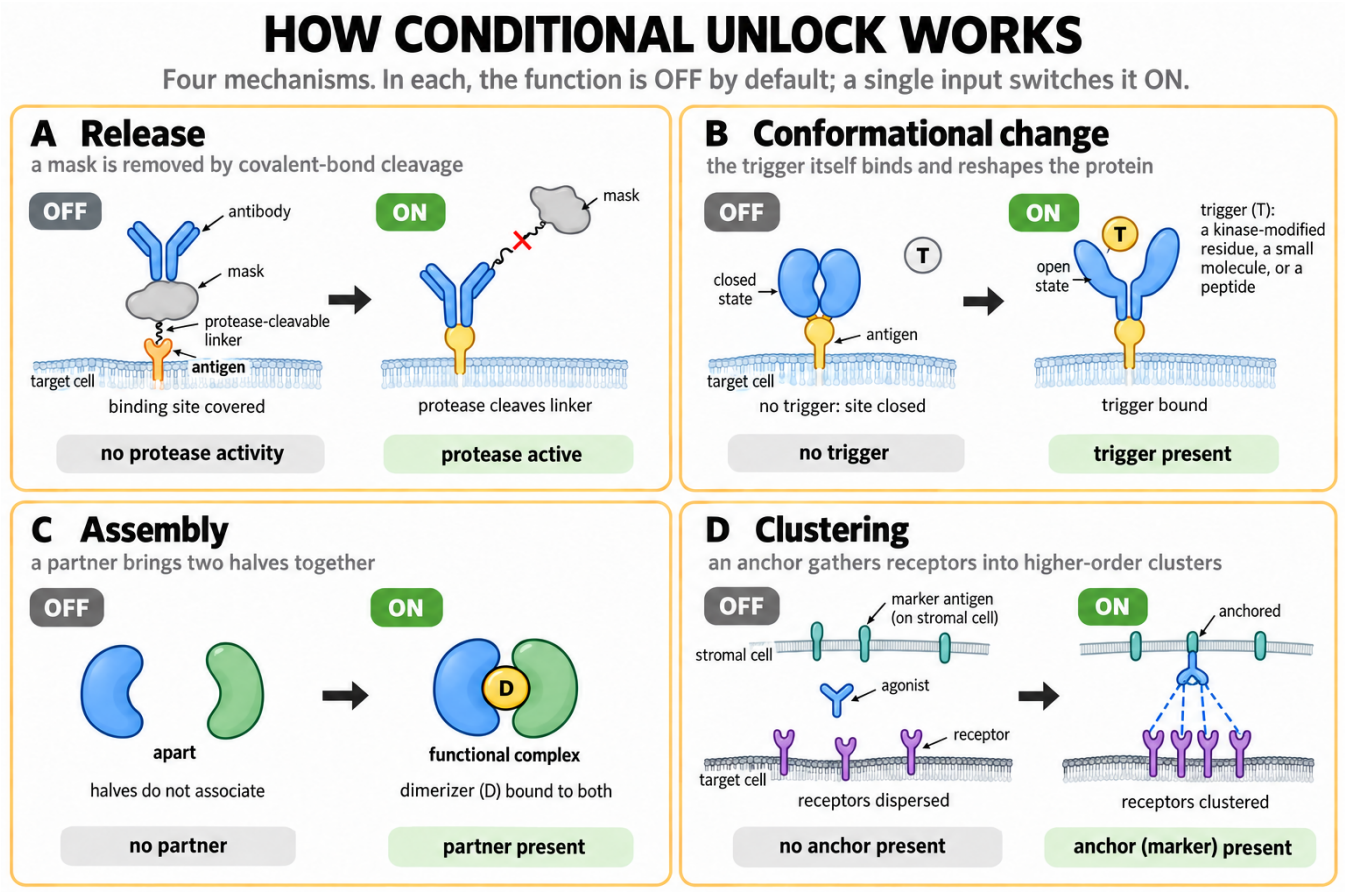


Figure 3: How Conditional Unlock works. Panel A: Release (a protease removes a mask by cleaving its linker). Panel B: Conformational change (the trigger binds and reshapes the protein). Panel C: Assembly (a partner brings two halves together). Panel D: Clustering (an anchor on a stromal cell gathers receptors on the target cell). In each panel, the function is OFF by default, and a single input switches it ON. Schematics are simplified and not to scale.

response. It also shows the continuity that exists within Conditional Unlock.

- A cleavable-mask design for a miniprotein binder [56].
- RELEASE [57] removes an ER-retention signal by protease cleavage. This permits secretion and surface display.
- PRCIS [58] is a protease-responsive, structurally repressed synthetic transcription factor. Its two DNA-binding domains are each held by a cleavable peptide, and full recovery needs both cleavages, so this review registers it as AND-like (Structural).
- A synpoptosis circuit [59] uses protease-controlled caspase-3/gasdermin effectors.

All of these belong to the same mechanism.

Sensor type does not determine the Release mechanism. HAP18 [25] is a hypoxia-activated antibody. It works through the same Release mechanism, removal of a PEG mask. However, its Sensor is not protease activity. It is hypoxia, sensed through azoreductase reduction of an azo bond. This is a physicochemical condition, and the Grammar records it as a Context.

We also confirmed scaffold diversity within this single mechanism. Different companies have implemented masked antibody/T-cell-engager/cytokine platforms on different scaffolds.

- Probody (CytomX): a single-arm peptide mask.
- SAFEbody/POWERbody (Adagene): a one- or two-arm peptide mask, plus Fc engineering.
- COBRA (Takeda/Maverick): removal of an inactivating single-chain fragment.

- TwoGATE (Revitope): reconstitution of two half-molecules, plus a PK switch.
- TRACTr (Janux): peptide masks found by phage display, with an albumin-binding domain appended to the mask [60]. PRO-XTEN (Vir/Amunix): XTEN-polymer masks.
- Anti-idiotypic mask: in a folate receptor 1 T-cell bispecific antibody, an anti-idiotypic anti-CD3 scFv serves as the mask, tethered by a protease-cleavable linker [61]. The mask is not a peptide.
- ProTriTAC (Harpoon): the anti-albumin domain of a TriTAC carries the mask. One proteolytic event removes the mask and the albumin-binding domain together [62].
- TriTAC-XR (Harpoon): a peptide mask is added to the N-terminus of a TriTAC. Proteases in systemic circulation activate it gradually, not at the tumor [63].

Among masked cytokines, the masking domain of XTX301 [64] is specifically confirmed in the primary literature. XTX301 is from Xilio, and its mask is derived from the IL-12 receptor β2 subunit. Werewolf's INDUKINE molecules, WTX-124 [65] and WTX-330 [66], use the same protease activation principle. Each carries an inactivation domain and a half-life extension domain, tethered to the cytokine by protease-cleavable linkers. In WTX-124, the inactivation domain is a Fab antibody, and the half-life extension domain is a single-domain antibody against albumin [65]. The sequences of the Fab and the linkers are proprietary.

The same combination appears in at least six molecular designs from different companies. The evidence reviewed here does not establish whether they arose independently or by transfer of the Probody concept. The combination is Sensor (Context: protease) × Gate (Conditional Unlock) × mechanism (Release).

Regarding scaffold frequency, the most frequently repeated basic form in this class is single-arm masking of one arm with a peptide mask. The masked arm is usually the CD3/effector arm. Examples are Probody, SAFEbody, the single-arm version of POWERbody, and the single-arm version of TRACTr.

Independent dual-arm masking is a later, more complex variant. Examples are XPAT, TwoGATE, ADG138, JANX008, and reportedly JANX014 and the VIR-5500 series of the same type. It is typically developed by the same company after it has established a single-arm version. Examples are Adagene's ADG152 → ADG138 and Janux's JANX007 → JANX008.

Steric occlusion by an XTEN polymer (XPAT, PRO-XTEN) is a physically distinct principle from peptide masking. Among the registered technologies, only these two use it, and both derive from Amunix.

Conformational change (covalent modification: phosphorylation-driven structural change)

A de novo-designed, kinase-driven protein switch [67] is triggered by a kinase such as Src or PKA. The kinase phosphorylates an interface residue within a Cage-Latch structure. This directly weakens binding energy at that residue. The latch is displaced, and the bundle opens.

ERK-phosphoGal4 [68] inserts a natural phosphorylation-recognition module into a Gal4 transcription-factor scaffold. The module is a WW domain paired with a phosphorylated substrate motif. ERK kinase activity is the trigger. It induces intramolecular binding of the WW domain. The resulting structural change permits Gal4 transcriptional activity.

Both examples mirror the protease-based systems above. An enzyme covalently modifies the protein, and the modification directly drives a structural change. In these cases the enzyme is a kinase that phosphorylates, and it does not cleave.

Conformational change (non-covalent: allosteric effects of small molecules and peptides)

Four examples differ from the covalent modifications above. In each, the trigger molecule itself binds non-covalently and reversibly, and this alters the structure.

- The Rip-3/Stitch-3 system [69] and the β-glycosidase W33G/indole system [70], [71]. Both use chemical rescue of structure: an exogenous small molecule fills a cavity and restores the structure.
- The INSRTR ON-switch [72]. It uses allosteric strain from an inserted peptide.

- A calmodulin-linker scFv [73]. A calmodulin-binding peptide induces a conformational change in VH/VL.

The Switch-Ig platform [26] uses an endogenous small molecule as the trigger. It restores antigen binding only in the presence of extracellular ATP. STA551 [74] is an anti-CD137/4-1BB agonist. ROSE12 [75] is an anti-CTLA-4-directed molecule. Both are built on this platform. ATP intercalates at the binding interface and induces a structural change. This places them mechanistically close to the non-covalent allosteric mechanisms of Rip-3/Stitch-3, β-glycosidase W33G/indole, and the calmodulin-linker scFv.

Assembly (Partner-induced association and complex formation, including oligomeric-state transitions)

Heterodimerization and complex formation.

- A chemically induced dimerization (CID) system [76] uses FKBP/FRB-type domains. They heterodimerize only when a synthetic dimerizer is present. This recruits a transcriptional activation domain to a promoter. It works as a reversible, dose-dependent on/off switch. A safety switch built on the same platform, iCasp9, has been given to pediatric patients in the clinic [77].
- A LITE switch [78] applies the same CID mechanism to antibody-complex formation. It has been demonstrated in three distinct therapeutic functions: tumor localization, extended cytokine half-life, and T-cell-engager activation. It is dose-reversible.
- A VH/VL small-molecule-gated Fv reassembly system [79] uses an antibody's own VH and VL variable domains. They associate to form a functional Fv only when a small-molecule ligand is present. The ligands include fluorescein, nicotine, and beta-estradiol. The fluorescein version has been used for in vivo gating of CAR-T and BiTE activity.

Oligomeric-state transitions. A de novo-designed hinge module [27] drives allosteric assembly. Three effects have been shown: effector-peptide-induced assembly, switching between dimeric states, and disassembly of a dihedral cage.

Assembly as a mechanism also appears in AND-like, where several Identity inputs together drive complex formation. Here, in contrast, a single Partner input alone drives it. The same mechanism can therefore be AND-like or Conditional Unlock, depending on the number of inputs that release the OFF state.

Clustering (higher-order receptor clustering through binding to a localization marker)

FAP-4-1BBL (RO7122290) [28], [29], [80] has a simple Sensor: the presence of FAP as a localization marker. It does not sense FAP's enzymatic activity. Its mechanism is higher-order clustering of the 4-1BB receptor, mediated by FAP-expressing stromal cells.

We call the domain that binds the localization marker an "anchor." It tethers the molecule to a specific tissue site, such as tumor stroma.

This is a fourth mechanism. It is distinct from Release and Conformational change. It does not "cut" a complex, and it does not "deform" one. It "gathers" one.

We also confirmed scaffold diversity within this mechanistic class. The same Sensor/Gate/mechanism combination is implemented on at least eight distinct scaffolds identified so far. The combination is localization marker × target-receptor agonism. The scaffolds are:

- an Fc fusion with a trimeric ligand (FAP-4-1BBL/RO7122290);
- a DuoBody controlled-Fab-arm-exchange IgG [81] (GEN1046 [30]);
- an IgG-scFv fusion (ABL503 [82], [83]);
- multivalent DARPin fusions [84] (MP0310 [85], MP0317 [86]);
- an antibody-Anticalin fusion [87] (PRS-343 [88]);
- a tetravalent mAb2/Fcab [89] (FS222 [90]);
- a common-light-chain IgG (MCLA-145 [91]);
- an scFv-type fusion (ABBV-428 [31]).

Regarding scaffold frequency, four of the nine molecules are conventional bispecific IgGs. They are GEN1046, ABL503, MCLA-145, and ABBV-428. They repurpose established antibody-engineering platforms: controlled Fab-arm exchange, common light chain, or simple scFv fusion.

The other four each rely on a novel domain platform held by a single company. They are the DARPins of MP0310 and MP0317, the Anticalin of PRS-343, and the Fcab/mAb2 of FS222. Among these nine, no platform is shared across companies. FAP-4-1BBL is likewise a singular example of its scaffold. Repurposing conventional IgG platforms is the most frequent single approach (four of nine). Each novel domain platform belongs to one company.

Which scaffold most reliably yields sharp Gate behavior remains a limited comparison (N = 3). Three examples explicitly report activity with and without the anchor.

- The Fc fusion with a trimeric ligand (FAP-4-1BBL) and the DuoBody-type IgG (GEN1046) both show activity below the detection limit without the anchor. This is sharp, binary behavior with direct head-to-head evidence.
- The direct scFv-type fusion (ABBV-428) shows graded, leaky behavior. The authors themselves report that activity is weaker but not abolished without the anchor.

Across Conditional Unlock. Conditional Unlock can be described with the same mechanism vocabulary as AND-like: Release, Conformational change, Assembly, and Clustering. Conformational change is further divided into covalent modification and non-covalent allostery. The two Gate values differ only in how many inputs are each independently needed to release the OFF state of the function. Sensors that bind the target or recruit an effector, such as the antigen and CD3 of a conventional T-cell engager, are not counted. The mechanism vocabulary itself is shared across Gate values, as the Assembly observation above confirms.

3 Discussion I: Engineering Difficulty, Gate Boundaries, and Distribution Patterns of Trigger Chemistry

Engineering difficulty: three AND-like designs were reached through adjustment

LINK CAR [10] and the P-cadherin × CDH17 antibody-drug conjugate [42] followed a similar path. Each started from a first design that responded partially to a single input alone. Each gained dependence on the combination of conditions only through repeated adjustment. The adjustments were mutation of a transmembrane region and deletion of a signaling-scaffold binding site, or the addition and re-selection of variants. AvidCAR [17] reached AND behavior through a different lever. It uses low-affinity binders, and substituting higher-affinity binders abolishes the behavior.

LINK CAR (Structural) and AvidCAR (Affinity-driven) make opposite choices about binding cooperativity. This is an observation from two designs, not a property of the sub-classes. In LINK CAR, to make the output close to a binary, the authors had to eliminate a leak on purpose. The leak came from homodimerization of the CD28 transmembrane domain, which is a form of binding cooperativity. They removed it by mutation (2CA).

The reason is as follows. Consider a simple combination of a LAT-CAR and an SLP-76-CAR. Antigen binding at one CAR then recruits the other CAR to the same synapse through homodimerization. Binding of a single antigen is enough to bring both intracellular domains together. In practice, this approximates an OR gate.

AvidCAR is Affinity-driven. It takes the opposite approach and deliberately exploits the same kind of cooperativity (avidity). It combines a low-affinity binding domain with a dimerizing hinge. The result is a continuous response tuned to antigen density.

The two designs sit at opposite ends of one design choice. One can eliminate binding cooperativity to build a binary AND-like output. Alternatively, one can exploit it to build a continuous response.

In these three examples, AND-like behavior did not emerge from the first design. It required optimization to acquire. The three are a subset of AND-like technologies. For many others, such as Hemibody, no prototype history is reported, so we cannot estimate how general this pattern is from our sample. We also have no comparison with Conditional Unlock designs, which may have needed adjustment too.

Preliminary observations on Gate boundaries

AND-like and Threshold-like. The selectivity of AvidCAR, the P-cadherin × CDH17 conjugate, JMB2403, and ISB 1442 is described as a continuous dependence on affinity, avidity, or expression level. Hernandez-Lopez [92] quantified ultrasensitivity with Hill coefficients. It is a useful reference point. The selectivity of these technologies is mechanistically close to that framework.

AND-like and OR-like. SUPRA CAR [93] is a second, independent example, alongside LINK CAR above. It shows that achieving AND-like as a Structural sub-class requires a deliberate structural intervention. Its AND-gate construct splits the signaling components across two separate zipCAR receptors. One carries CD3ζ, and the other carries a co-stimulatory domain such as CD28. Full signaling is reconstituted only when both corresponding zipFv adapters are present. LINK CAR made the same design choice when it split LAT and SLP-76.

An OR gate has also been built on the same zipCAR/zipFv platform (targeting FLT3 × CD33 [94]). Either zipFv alone is sufficient to kill the corresponding single-antigen-positive cells. That the same platform can build an OR gate suggests that OR-like behavior results when the split is absent. A receptor with one antigen-binding domain responds to its own antigen alone. Two such receptors combined remain OR-like unless a structural dependence is added. This is our interpretation. In the two examples, LINK CAR and SUPRA CAR, AND behavior required a deliberate structural intervention: splitting the signaling components.

AND-like and NOT-like. Co-LOCKR [14] and CIPHR [34] both achieve NOT-like. Co-LOCKR does so by competitive sequestration of the Key by a decoy domain. CIPHR does so by competitive dissociation of an output homodimer by an input protein. Each platform also hosts an AND-like construct.

NOT-like is outside the Gate scope of this review. Nevertheless, the pairing is worth recording. The same physical principle, competitive binding, gives AND-like when combined with a colocalization condition. It gives NOT-like when combined with a sequestration or dissociation condition. A Gate value thus depends on how Sensor and mechanism are combined.

Scaffold diversity within the same mechanism

This section builds on the scaffold-diversity observations above. They concern two families: masked antibody/T-cell-engager/cytokine molecules, and localization-marker-anchored 4-1BB/CD40 agonists.

The same Sensor × Gate × mechanism combination appears in six to eight different scaffolds. The combinations are Context × Conditional Unlock × Release, and Context × Conditional Unlock × Clustering. Gate is the logic of functional permission. The mechanism is the physical way the Gate is implemented. The same pairing recurs across different scaffolds. This shows that the underlying design principle, not just its molecular execution, recurs across companies. It does not show that the recurrences were independent.

The earlier discussion of engineering difficulty concerned the design process of a single technology. Acquiring AND-like behavior within a given scaffold requires iterative adjustment. This section makes a higher-order observation about the technology landscape itself: the same underlying mechanism recurs across scaffolds many times.

The following is a hypothesis based on a limited sample. The most frequently adopted scaffold does not necessarily achieve the sharpest gating. The common scaffolds are a conventional bispecific IgG and single-arm peptide masking.

FAP-4-1BBL [28], [29] is an Fc fusion with a trimeric ligand, and it is a singular example. GEN1046 [30] is a DuoBody-type IgG, one of the more widely adopted formats. Both show sharp, binary behavior. Direct head-to-head evidence shows zero activity without the anchor.

ABBV-428 [31] is a direct scFv-type fusion. It also repurposes a conventional IgG format. However, its own authors report graded, leaky behavior. We do not claim a general correlation from a comparison of only three examples.

For FAP-4-1BBL, one plausible reason for the sharp on/off ratio is its design. Its trimeric ligand is an Fc fusion presenting three copies of 4-1BBL. This deliberately builds in the higher-order clustering that 4-1BB, a TNFRSF-family receptor, requires for signaling [95].

Why protease-mediated triggers predominate within the Release mechanism for Conditional Unlock

This review registers 38 Conditional Unlock technologies, counted by principal Gate value. We classified them by mechanism. Release by irreversible covalent-bond cleavage is the largest group. The great majority of these depend on proteases. These counts and distributions describe the registered technologies. They are not a census of the literature. The therapeutic platforms include Probody, masked cytokines, COBRA, and the single-mask forms of TRACTr, among others. The other registered technologies include RELEASE, a synpoptosis circuit, and PAS:LF [96], among others.

A few examples are counterexamples supported by evidence. They achieve the same Release mechanism with a non-protease trigger.

- Photobody [97] removes a photolabile cage with UV light.
- HAP18 reduces an azo bond under hypoxia.
- Sw-IM [98] cleaves a linker in response to reduction potential or acidic pH.

All three share the mechanism of the protease-based systems: irreversible covalent-bond cleavage.

This observation supports two points.

First, Photobody, HAP18, and Sw-IM suggest that the core design principle of Conditional Unlock is not specific to proteases. It is irreversible covalent-bond cleavage itself. Irreversible cleavage has a pharmacological advantage. The activated state can be maintained after the molecule reaches the target tissue, even if contact with the trigger was brief. Reversible mechanisms (Conformational change, Assembly, Clustering) differ. Their activated state lasts only while the trigger is present.

Second, irreversibility alone does not explain why protease-based systems so greatly outnumber nonenzymatic irreversible Release mechanisms (light, redox). We suggest at least two contingent factors.

(a) **Availability as an endogenous trigger.** Light and redox triggers typically need external, artificial stimulation. Tumor-associated protease activity exists inside the tumor microenvironment. It can sense the disease state without external manipulation.

(b) **Engineering maturity and technology transfer.** ADC and prodrug linker development has built up a catalog of protease-cleavable peptide sequences [45]. These are short, genetically encodable sequences. They transfer easily to new targets.

We interpret the predominance of proteases as the joint outcome of a general principle and these two contingent factors. The general principle is that irreversible Release favors sustained activation. We do not read it as evidence that a protease-rich environment is specially privileged as a Conditional Unlock mechanism.

4 Trigger Recognition Architecture: Design Tactics, Execution Platforms, and Principled Extensibility

The preceding discussion covered distribution patterns, boundaries, and engineering difficulty, as they emerged from the review above. This section examines the protein architecture of individual technologies directly.

It has three parts. First, we classify design tactics by the mechanism that recognizes the trigger and satisfies the condition. Second, we examine the computational and AI platforms that carry out these tactics. Third, we integrate both to address a higher-order question: how far can trigger chemistry be extended, in principle?

Tactical types for affinity setting

The Affinity-driven sub-class and competitive mask-release designs share a design problem. Competitive mask-release designs include Co-LOCKR, CIPHR, and Switch-DARPin. The problem is this: which binding energy should be set below the threshold needed for function on its own? The effective affinity rises only when several conditions are met together. That rise pushes the interaction over the threshold. We classify this as a set of reusable tactics.

Type 1 (deliberate sub-threshold affinity). AvidCAR [17] deliberately selects a low-affinity binding domain (Kd 877 nM). At this level the domain does not function as a monomer. Binding lifetime exceeds the functional threshold only through the avidity that dimerization generates. A kinetic-proofreading model [99] predicts this dependence quantitatively and confirms it. A direct control shows that the dependence disappears at higher affinity (Kd 13 and 125 nM).

Type 2 (asymmetric arrangement: division of labor between anchor and trigger arms). Three designs pair a high-affinity anchor arm with a weakened trigger arm.

- Evobody [43] pairs a BTN3A arm (the trigger) with a high-affinity FOLR1 arm (the anchor). The BTN3A affinity is deliberately reduced by germline reversion.
- JMB2403 [18] pairs a high-affinity PD-1 arm (the anchor) with an intrinsically weak IL-2R$\beta\gamma$ module (the trigger).
- ISB 1442 [20], [21] pairs a low-affinity CD47 arm (the trigger) with a biparatopic, high-affinity CD38 arm (the anchor). The CD47 arm is deliberately low in affinity, so that it blocks CD47 mainly on cells that carry CD38 (a "dock and block" design).

All three leave the anchor arm at high affinity. They weaken only the trigger arm, below the functional threshold.

Type 3 (empirical panel screening). The P-cadherin $\times$ CDH17 antibody-drug conjugate [42] screened 24 anti-CDH17 IgGs by their monovalent/bivalent potency ratio. It then used an alanine-scanning-mutant panel of the other arm's CDR3. From this panel it selected variants that lost single-antigen activity but kept double-positive activity. Types 1 and 2 aim for a target Kd by computation. This tactic finds the target Kd empirically, by screening a mutant panel.

Type 4 (avidity generation via added binding valency). ISB 1442's biparatopic CD38 arm has two epitopes [20], [21]. It raises local Fc density and avidity by increasing the number of epitopes. It does not raise the affinity of a single epitope. This tactic is a substitute when fine-tuning the Kd of a single domain would cost expression, manufacturability, or specificity.

Type 5 (reversal of effective affinity between intramolecular self-restraint and intermolecular trigger binding). Three designs use this tactic.

- Co-LOCKR [14] keeps its Latch bound within the same chain. This intramolecular binding has high effective affinity, and the default state is OFF. The Latch is released only when Key binding outcompetes it. That happens when antigen co-localization raises the two-dimensional avidity of the Key. A third component, a Decoy, provides a safety mechanism. It runs the competition in reverse by sequestering the Key.
- CIPHR [34] dissociates the homodimer of an output protein (default ON) with a competing input protein.
- SPOC [32] is discussed here as a single-input construct, although the platform also implements AND-like logic. It shares the same design problem. Split fragments that bound too strongly could not be held OFF by coiled-coil autoinhibition alone, and this caused leakage. To address this, a catalytically dead protease fragment was fused to the autoinhibitory side. This equalized the interaction energies of the ON and OFF states. Mutations at non-contact residues then narrowed the stability of the coiled-coil into a tight window. Inside this window, autoinhibition wins by default but permits displacement after cleavage.

This "energy window" concept differs from the avidity-dependent Types 1-4. It appears to be a general design problem, independent of Gate value. MP0632 (Switch-DARPin) [15] is plausibly analogous to Type 5.

Challenges and room for extension. Types 1-4 are avidity-dependent. All of them work only if the trigger (the antigen) is presented multivalently in the same region of a cell surface. The reason is physical. Several weak interactions concentrated in one place push the effective affinity over a threshold. This itself requires multivalent presentation of the trigger. These tactics therefore share a structural constraint. In principle they cannot turn the response to a single, monomeric, freely diffusing soluble trigger into a binary one.

Type 5 is different. It is driven by a one-to-one competitive binding equilibrium (Cage-Key, or homodimer versus competing input). It does not require multivalent presentation of the trigger. A single monomeric soluble protein or peptide can bind and outcompete the intramolecular self-restraint on its own. This holds for a single-input unlock, such as a soluble Key acting on LOCKR. Co-LOCKR, used as an AND-like gate, is designed differently. Its Latch is released only when two antigens co-localize on one cell surface and raise the two-dimensional avidity of the Key. A soluble monomeric Key that beat the Latch on its own would break the AND. As an AND-like gate, therefore, Co-LOCKR shares the dependence on trigger presentation of Types 1-4.

The two groups fall into different Gate categories. The Affinity-driven sub-class (Types 1-4) mainly uses two-dimensional binding-valency increase, that is, avidity. Competitive mask-release designs (Type 5) use the relative reversal of intramolecular and intermolecular effective affinity. However, both solve the same engineering problem: deliberately setting a sub-threshold binding energy that is exceeded only when several conditions are met together. They do it through different physical implementations.

The two are not equivalent in how they depend on trigger presentation: only Type 5, as a single-input unlock, can in principle extend to a monomeric, soluble trigger.

Tactical types for engineering conformational-change mechanisms

We now classify the structural design tactics for four kinds of design: LOCKR-family latch-key systems, covalent-modification-driven conformational change, non-covalent allosteric mechanisms, and hinge-module-driven oligomeric-state transitions. Affinity and energy-window tuning is covered under Type 5 above. This section focuses on the structural architectures that produce conformational change itself.

Type A (latch-key competitive-displacement architecture). LOCKR [100] encodes a functional motif on a mobile "Latch" helix. The motif can be a binding, degradation, or nuclear-export signal, for example. By default the Latch binds a Cage surface within the same molecule and hides the motif. An external "Key" peptide competes with the Latch for the same Cage interface. Only when the Key displaces the Latch is the motif exposed. This yields up to 40-fold activation.

A related construct from the same paper [67] (pGFP-4Y, pDIA-4Y, pGFP-4S) uses a Cage-Latch architecture without a separate Key: kinase-mediated phosphorylation of the Latch itself directly weakens its binding to the Cage, displacing it.

We see two design advantages in this architecture. First, the Cage-Key interface can be varied independently by sequence. This creates several orthogonal (mutually non-interfering) switches. The functional motif can also be encoded on the Latch independently of the Cage-Key pairing. The architecture is therefore a highly reusable template that transfers easily to new outputs. Second, the Cage-Key interface is a novel, computationally designed protein-protein interaction. It is not a repurposed natural recognition domain. It therefore avoids the structural constraint shared by Types B-D below. Those types can accommodate only a short linear motif or a small molecule in a binding pocket or groove.

Type B (grafting of a natural allosteric recognition module). ERK-phosphoGal4 [68] inserts a natural phosphorylation-recognition module into a Gal4 scaffold. The module is a WW domain paired with a phosphorylated substrate motif. ERK phosphorylation induces intramolecular WW-domain binding. This drives an allosteric conformational change.

A calmodulin-linker scFv [73] replaces the inert structural linker between the VH and VL of an scFv. The new linker is a natural Ca2+-dependent allosteric module, calmodulin. Its structural change propagates mechanically. It modulates the relative VH/VL orientation and the shape of the paratope.

The evidence indicates one design constraint. Switchable variants were obtained for 5 of 6 scFvs tested with the calmodulin-linker design. No switchable variant could be obtained for an anti-FITC scFv with extremely high intrinsic affinity (270 fM). This suggests that the mechanism fails when the structural rigidity or affinity of the target domain exceeds the structural change the inserted module can transmit.

Type C (chemical rescue via cavity complementation). Two systems use this tactic. The

Rip-3/Stitch-3 system [69] has a cavity-forming mutation at the VH-VL interface. The small molecule Stitch-3 restores it. The β-glycosidase W33G/indole system has a cavity-forming mutation that removes a buried active-site tryptophan. Indole restores it in a dose-dependent way, with up to roughly 780-fold recovery in kcat/Km [70].

Both systems deliberately disrupt structure and activity by removing a buried, bulky side chain. Both restore near-wild-type structure and function only when an exogenous small molecule occupies the resulting cavity.

The design insights are as follows.

- The choice of which side chain to remove affects behavior.
- Screening a panel of candidate small-molecule analogs identifies the best complementing agent. In the β-glycosidase system, 13 of 15 indole analogs partially rescued activity [71].
- Adding a cavity-enlarging mutation permits specific rescue by a larger synthetic molecule.
- The behavior tends toward a continuous "rheostat" rather than a binary switch. This contrasts with Type A.
- ROSE12 [75] appears to be a related form of cavity- and interface-dependent allostery. In ROSE12, ATP intercalates at a binding interface and induces a structural change.

There is also a methodological limitation. A related series for the β-glycosidase system used three unrelated *E. coli* homodimeric proteins with 14 Trp→Gly mutations [101]. Rescue success correlated with the Rosetta-estimated stability change, not with interfacial energy. The original authors therefore argue that the effect is better explained as recovery of overall folding stability than as a localized conformational change.

We note this as a limitation of the Conformational change label itself. Without structural evidence, it can be difficult to distinguish this mechanism from alternative explanations.

Type D (an inserted allosteric lever at a permissive site). INSRTR [72] inserts a coiled-coil-forming peptide at a permissive loop about 2 nm from the active site, not at the active site itself. Binding of a co-expressed partner peptide (REG) suppresses activity. The degree of suppression is proportional to the affinity of the coiled-coil pair. This is a continuously tunable "dial," not a binary switch.

An inverted, ON-switch version of the same module has also been demonstrated. The same inserted-lever architecture can therefore work as an ON-switch or an OFF-switch, depending only on the chosen topology.

Type E (hinge-module engineering for association-state transitions). A de novo-designed hinge module [27] drives allosteric assembly. A computationally designed, rigid "hinge" module is inserted between oligomer-forming subunits. Binding of an effector peptide then triggers a transition between oligomeric states, for example from a C3 ring to a C4 ring. Cooperative responses with Hill coefficients of 2.7-3.9 have also been observed.

We interpret the design insight as follows. Control "state X rings" were designed without built-in strain. None of the 9 soluble ones (24 designed) switched. Designs with deliberately introduced strain switched in 10 of the 17 soluble cases (26 attempted). In this study, switching required the default associated state to carry structural strain. This is consistent with the interpretation that allosteric switching requires the default state to lie "close" to the transition. It is, however, the result of a single study, with 9 soluble controls. Whether it applies to the designs of Types A-D has not been tested, and remains a hypothesis.

Table 2 compares the five types.

Summary. The five types organize at least three distinct structural ideas within the single mechanism label of Conformational change.

- Competition for a binding site (Type A, with Type B as a variant).
- Small-molecule complementation of a deliberately disrupted structure (Type C).
- Allosteric propagation from an inserted lever at a permissive site (Type D).

In addition, a single study (Type E) shows the idea of engineering a strained ground state that is poised for transition. Its generalization to the other types has not been tested.

The preceding section dealt with a quantitative tactic: how weak or strong to make a binding interaction. This section deals with a topological one: where, and

Table 2: Structural design tactics for conformational change. Values are those reported by the original authors.

Type	Principle	Examples	Trigger	Design findings	Constraint or limitation
A	Latch-key competitive displacement	LOCKR [100]; caged CD3ζ ITAM switch [67]; pGFP-4Y/pDIA-4Y/pGFP-4S [67]	Key peptide; key peptide; kinase phosphorylation	Up to 40-fold activation; orthogonal Cage-Key pairs; the motif is encoded on the Latch independently of the pairing	Cage-Key is a designed protein-protein interaction, so it is not limited to short motifs
B	Grafting a natural allosteric module	ERK-phosphoGal4 [68]; calmodulin-linker scFv [73]	ERK phosphorylation; Ca2+ and a calmodulin-binding peptide	Switchable variants for 5 of 6 scFvs tested	No switchable variant for an anti-FITC scFv of extremely high affinity (270 fM); trigger limited to a short motif or small molecule
C	Chemical rescue by cavity complementation	Rip-3/Stitch-3 [69]; β-glycosidase W33G/indole [70]	Small molecule that fills the cavity	Up to roughly 780-fold recovery; 13 of 15 indole analogs partially rescued activity [71]; rheostat-like behavior	Rescue may reflect recovery of folding stability rather than a localized conformational change [101]
D	Inserted allosteric lever at a permissive site	INSRTR [72]	Partner peptide (REG)	Suppression proportional to coiled-coil affinity; ON- or OFF-switch depending on topology	Trigger limited to a short peptide
E	Hinge-module engineering for association-state transitions	De novo hinge module [27]	Effector peptide	10 of 17 soluble strained designs switched (26 attempted), with Hill coefficients of 2.7-3.9; unstrained controls never switched	In this study, a strained default state was required

how, to build competition, propagation, or strain into a structure.

There is also a challenge, and room for extension. Type B, Type D, and part of Type C (the ATP pocket of ROSE12) all recognize their trigger through a short linear motif, a small molecule, or an ion. It is held in an evolutionarily or artificially optimized groove or pocket. This shape is a structural constraint that engineering effort cannot extend. In principle it cannot hold a folded protein domain (several to several tens of kDa).

Type A (Latch-Key) is not subject to this constraint. Its Cage-Key interface is itself a novel, computationally designed protein-protein interaction. We develop this contrast further below.

Computational and AI design platforms that execute affinity-setting and conformational-change strategies

The preceding sections classified which design patterns are being pursued. This section asks a narrower question. Within those patterns, what do computational and AI design platforms replace or speed up?

Precision tuning of a targeted energy window by sequence- and interface-design models. Several tools serve the same role. They are Rosetta ddG calculations [102], FoldX [103], inverse-folding models (ProteinMPNN [104] and LigandMPNN [105]), and protein language models (ESM [106]). ESM uses pseudo-log-likelihood to filter variants for plausibility. All of them computationally narrow down which variants of an existing scaffold fall within a targeted range of binding energy.

This role matches two requirements. One is the requirement in Types 1 and 2 above: a Kd that gives no function on its own but falls within a specific range. The other is the requirement in Types A, C, and E above: an energy window in which the default state holds only a slight advantage but can be reversed by an external trigger. MASS [107] is a different kind of tool. It models how a multivalent binder engages surface antigens, and predicts how affinity and valency set its selectivity for antigen combinations and densities. Its predicted selectivity matched measured EC50 ratios on engineered cell lines. It serves the requirement of Types 1 and 2 directly.

Computational pre-filtering also replaces part of the exhaustive alanine-scanning and mutant-panel screening used in Type 3 (the P-cadherin × CDH17 antibody-drug conjugate). It reduces the scope of that work.

De novo generation of new binding domains and scaffolds by deep learning — and why

de novo generation is necessary. The tactics above demand properties that natural antibody- and scaffold-discovery pipelines were never designed to optimize. These pipelines include phage/yeast display and hybridoma generation, followed by affinity maturation. We see three reasons.

First, Types 1 and 2 need deliberately weak binding domains within a specific Kd window. This runs against the direction of affinity maturation, which aims for the strongest binder possible. An alternative is to design for a target ddG/Kd range from the start. One can parameterize the number of interfacial hotspot residues and the buried surface area directly. Whether this yields a working window more reliably than simpler methods, such as the selection used for AvidCAR or the germline reversion used for Evobody, is not shown in this review.

Second, some designs need several binding domains that do not cross-react with one another. LOCKR needs orthogonal Cage-Key pairs for AND-gate design. Discovery-based methods have difficulty producing such sets systematically. They depend on the incidental paratope shapes that the immune system or a display library happens to return against a single antigen. De novo design can generate new interfacial sequences by computation. In principle, it can produce many orthogonal binding pairs.

Third, some self-restraining architectures have no natural counterpart. Cage/Latch itself is one (Types A and E). We infer that such architectures are hard to reach by repurposing a known natural domain or by screening an antibody library. Designing a novel fold is the practical route.

RFdiffusion [108] generates backbones specific to a target epitope. End-to-end hallucination pipelines are also available, such as BindCraft [109], boltzgen [110], and Mosaic [111]. These tools can in principle address the three requirements. This review cites no gated molecule designed with them. The statement is a prospect, not a demonstrated result.

The immaturity of generative AI specifically for conformational-change design. The two roles above are tasks of building or tuning a single structure, either an existing scaffold or a novel one. Conformational change is different. Its essence is bistability: the protein is locked into one state by default and switches to the other only when the trigger is present. AI platforms that design bistability itself are much less mature. Rosetta ddG calculation and de novo binder generation are more mature for affinity tuning.

Current leading generative and predictive models are RFdiffusion [108], AlphaFold2/3 [112], [113], and ESM [106]. They are optimized to output "one dominant structure for one sequence." In the examples reviewed here, bistable designs, such as the hinge-module-driven oligomeric-state transitions above, do not rely on deep generative models. They rely on physics-based computational design. Large numbers of candidates are then screened experimentally for solubility and switching. In the hinge-module work, 10 of the 17 soluble designs switched (26 attempted).

One approach is partly relevant: conformational sampling. It stochastically samples the multiple conformations of a natural protein, for example through MSA subsampling in AlphaFold2 [114]. It is not a tool for generating a novel bistable switch. It is a verification-oriented method that repurposes an existing prediction model. Nevertheless, it could in principle help design proposals like Type E (association-state transitions driven by hinges and strain). It could check in silico, beforehand, whether a candidate sequence can plausibly adopt both states.

Overall, computational and AI design platforms may replace or accelerate two elements of the tactics above. One is precision narrowing of an energy window on an existing scaffold. The other is de novo generation, where properties are difficult to reach through natural discovery pipelines. These properties are deliberate weak affinity, mass production of orthogonal binding pairs, and novel self-restraining folds.

Progress is uneven between the two design problems. For affinity tuning, a role for generative AI is plausible on both fronts. Direct demonstrations in gate design are not covered in this review. For conformational-change design, AI tools aimed at bistability itself remain immature. Dependence on physics-based computation and experimental screening remains comparatively high.

Table 3 summarizes the roles.

Table 3: Roles of computational and AI design platforms.

Role	Tools	What they do	Related types
Energy-window tuning on an existing scaffold	Rosetta ddG [102]; FoldX [103]; ProteinMPNN [104]; LigandMPNN [105]; ESM [106]	Narrow down which variants fall within a targeted range of binding energy; ESM filters variants by pseudo-log-likelihood	Types 1, 2, A, C, E; reduces the screening needed for Type 3
De novo generation of binding domains and scaffolds	RFdiffusion [108]; BindCraft [109]; boltzgen [110]; Mosaic [111]	Can in principle be used to design deliberately weak binders, sets of orthogonal binding pairs, and novel self-restraining folds	Types 1, 2, A, E
Conformational sampling (verification)	MSA subsampling in AlphaFold2 [114]	Sample the conformations of a natural protein; check in silico whether a candidate can adopt both states	Type E (potential use)
Design of bistability itself	None mature	Current models (RFdiffusion, AlphaFold2/3 [112], [113], ESM) output one dominant structure per sequence; bistable designs still rely on physics-based design and experimental screening	Conformational-change designs (Types A-E)

The principled extensibility of trigger chemistry to protein-type ligands, for reversible Conditional Unlock/AND-like

The two preceding sections reached a parallel contrast on their own. It concerns how far trigger presentation and trigger recognition can be extended.

- The affinity-setting section found that avidity-dependent tactics are limited to multivalent, surface-presented triggers. Competitive-displacement tactics can extend to monomeric, soluble triggers.
- The conformational-change section found that designs repurposing natural motif or small-molecule recognition domains are constrained by the shape of a binding pocket or groove. Cage-Key competitive-displacement designs are not.

This section combines the two contrasts along a single axis: trigger chemistry, with particular attention to extension toward protein-type ligands. We state explicitly that this argument concerns the physical constraints imposed by the binding mode itself. It does not concern how far any individual technology has been validated experimentally.

The challenge. Trigger recognition based on a linear motif or a small-molecule pocket is built around a small trigger. As designed, it does not accommodate a protein-type ligand. Each of the following recognizes a small trigger.

- ERK-phosphoGal4 (a WW domain) accommodates a short, proline-rich phosphorylation motif.
- A calmodulin-linker scFv accommodates a single M13-like amphipathic helix.
- INSRTR accommodates a short coiled-coil-forming peptide.
- ROSE12, the Switch-Ig (ATP-switch) platform, and a VH/VL small-molecule-gated Fv reassembly system each accommodate small molecules of a few hundred daltons.

In principle, a trigger-recognition module can be redirected to a different natural factor. Rewiring ERK-phosphoGal4 from ERK to JNK demonstrated this directly [68]. However, all of these designs share the more basic structural constraint noted above: the replacement trigger must still be a short motif or a small molecule, not a folded protein domain.

Avidity-dependent Types 1-4 have a constraint noted above: thresholding is possible only for a multivalent, surface-presented trigger. It is the same kind of structural limitation, this time on how the trigger must be presented. It has the same root cause.

The most extensible architecture in principle. Architectures whose trigger-recognition binding mode is itself a protein-protein interaction are free of this constraint.

Consider the association logic of the CID system and the LITE switch (Partner-induced Assembly of the CID type). Two half-molecules associate only when a given Partner molecule is present. In principle, this logic is chemically neutral to the identity of the Partner. The Partner can be a small molecule or a folded protein. Natural biology routinely uses exactly this logic. A cytokine or growth factor bridges two receptor extracellular domains and induces heterodimerization [115].

The FKBP/FRB-type domains in these systems are only one implementation, built around a synthetic dimerizer. We infer that replacing this association-inducing domain with a pair of natural receptor extracellular domains would extend the trigger to a natural protein factor. The same Assembly architecture would be preserved. This remains untested.

LOCKR (Cage-Latch-Key, competitive displacement), used as a single-input unlock, belongs to the same group. Its Cage-Key interface is the same kind of designed protein-protein interaction noted above. The AND-like behavior of Co-LOCKR depends on co-localization on a cell surface, so the extensibility argument is limited to its single-input form.

De novo protein design itself assumes only that a binding interface can be designed against an arbitrary target surface. The partner can be a short peptide or a larger folded protein. Unlike ERK-phosphoGal4 and the calmodulin-linker scFv, which repurpose natural recognition domains, it has no built-in limit to binding only a short linear motif.

Conclusion. Trigger-recognition architectures for reversible Conditional Unlock/AND-like fall into at least two groups by binding mode.

The first group is structurally constrained by how the trigger is presented. The trigger is either a short motif or small molecule, or a multivalent, surface-presented antigen. This group has two kinds of design.

- Natural motif/small-molecule-recognition-domain designs: ERK-phosphoGal4, the calmodulin-linker scFv, INSRTR, ROSE12/Switch-Ig, and the VH/VL small-molecule-gated Fv reassembly system.
- Avidity-dependent designs: AvidCAR, Evobody, the P-cadherin × CDH17 antibody-drug conjugate, JMB2403, and ISB 1442.

The second group has a binding mode that is itself a protein-protein interaction. It includes the Cage-Key of LOCKR (as a single-input unlock), CIPHR, and the Partner-induced Assembly of the CID system and LITE switch. Their trigger-recognition sites were designed around a one-to-one interaction with a protein-type ligand from the start. They have no structural barrier to extension toward a monomeric, soluble protein ligand. Co-LOCKR and Switch-DARPin (MP0632), used as AND-like gates, have two cell-surface antigens as the trigger and depend on co-localization. For them, the extensibility of this group reaches only the single-input form. The trigger of SPOC is a protease, which is a protein, but its cleavage is irreversible. We therefore do not place it in this section, which concerns reversible designs.

Figure 4 summarizes the two groups.

This distinction concerns an architectural property: extensibility to different trigger chemistries and presentation modes. It says nothing about which group has been validated more thoroughly in experiments. We treat it as an axis independent of practical requirements such as tumor selectivity, discussed above in connection with protease predominance.

5 Conclusion

This review covered 70 technologies. Of these, 67 have AND-like or Conditional Unlock as the principal Gate value. These two Gate values in the Protein Engineering Grammar are the conjunctive gates. The other three are boundary cases, whose principal Gate value is NOT-like, uncertain, or Direct. We organized a map of mechanistic diversity across modalities.

We documented the degree of specificity and activity that individual technologies achieved. We also documented the iterative design process behind it. In several designs, the initial prototype was leaky. AND behavior was reached only through adjustment. We also recorded preliminary observations on Gate boundaries relative to Threshold-like, OR-like, and NOT-like.

We then classified the protein architecture of trigger-recognition modules into reusable design tactics. There are five types for affinity setting and five types for conformational change.

We examined the current role of computational and AI design platforms in carrying out these tactics. They could be applied to fine-tuning energy windows and to de novo generation, although this review cites no gated molecule designed in this way. Their ability to design bistability itself is still immature.

Finally, we showed that trigger-recognition architectures fall into two groups by binding mode. One group is structurally constrained by how the trigger is presented. In the other, the binding mode

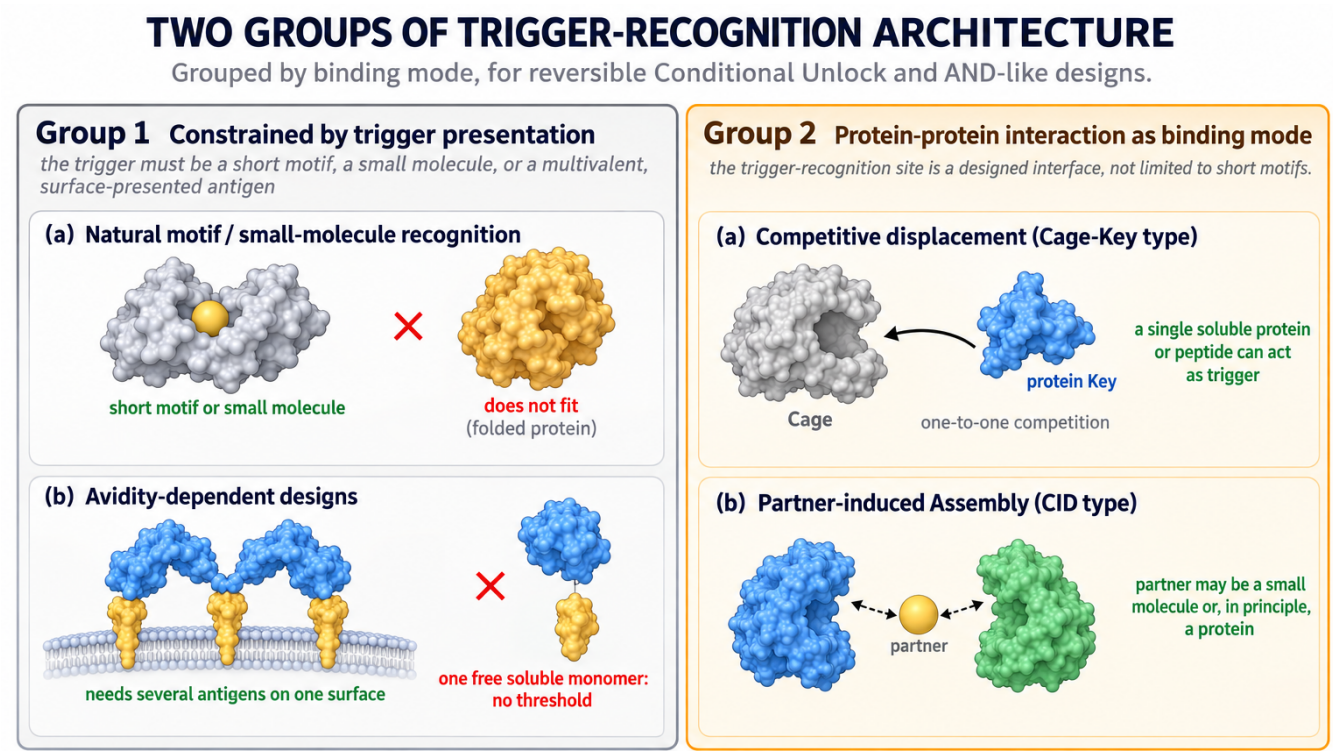


Figure 4: Two groups of trigger-recognition architecture. Group 1 is structurally constrained by how the trigger is presented: a short motif or small molecule in a groove or pocket, or a multivalent, surface-presented antigen. In Group 2 the binding mode is itself a protein-protein interaction, so a single soluble protein can in principle act as the trigger. The contrast concerns the physical constraint imposed by the binding mode. It does not rank the groups by how far each design has been validated experimentally.

is itself a protein-protein interaction. This second group therefore has a principled extensibility to trigger chemistry.

This depth is specific to a review narrowed to the conjunctive gates. It was not part of the broader overview of molecular implementation given in the prior Grammar paper [1].

References

[1] K. Saka, "Protein Engineering Grammar: From Molecular Formats to Programmable Medicines." Zenodo, 2026. doi: 10.5281/zenodo.22868586.

[2] L. R. Desnoyers *et al.*, "Tumor-Specific Activation of an EGFR-Targeting Probody Enhances Therapeutic Index," *Science Translational Medicine*, vol. 5, 2013, doi: 10.1126/scitranslmed.3006682.

[3] A. Banaszek *et al.*, "On-target restoration of a split T cell-engaging antibody for precision immunotherapy," *Nature Communications*, vol. 10, 2019, doi: 10.1038/s41467-019-13196-0.

[4] S. C. Oostindie *et al.*, "Logic-gated antibody pairs that selectively act on cells co-expressing two antigens," *Nature Biotechnology*, vol. 40, 2022, doi: 10.1038/s41587-022-01384-1.

[5] R. A. Morgan, J. C. Yang, M. Kitano, M. E. Dudley, C. M. Laurencot, and S. A. Rosenberg, "Case Report of a Serious Adverse Event Following the Administration of T Cells Transduced With a Chimeric Antigen Receptor Recognizing ERBB2," *Molecular Therapy*, vol. 18, 2010, doi: 10.1038/mt.2010.24.

[6] O. Nolan-Stevaux and R. Smith, "Logic-gated and contextual control of immunotherapy for solid tumors: contrasting multi-specific T cell engagers and CAR-T cell therapies," *Frontiers in Immunology*, vol. 15, 2024, doi: 10.3389/fimmu.2024.1490911.

[7] N. Kim and I. Kwon, "Conditionally Activatable Antibody Platforms: Mechanisms, Modalities, and Clinical Translation Potential," *BioDrugs*, vol. 40, 2025, doi: 10.1007/s40259-025-00752-y.

[8] E. M. Bressler, S. Adams, R. Liu, Y. L. Colson, W. W. Wong, and M. W. Grinstaff, "Boolean logic in synthetic biology and biomaterials: Towards living materials in mammalian cell therapeutics," *Clinical and Translational Medicine*, vol. 13, 2023, doi: 10.1002/ctm2.1244.

[9] M. Geis *et al.*, "Combinatorial targeting of multiple myeloma by complementing T cell engaging antibody fragments," *Communications Biology*, vol. 4, 2021, doi: 10.1038/s42003-020-01558-0.

[10] A. M. Tousley *et al.*, "Co-opting signalling molecules enables logic-gated control of CAR T cells," *Nature*, vol. 615, 2023, doi: 10.1038/s41586-023-05778-2.

[11] A. Quijano-Rubio *et al.*, "A split, conditionally active mimetic of IL-2 reduces the toxicity of systemic cytokine therapy," *Nature Biotechnology*, vol. 41, 2023, doi: 10.1038/s41587-022-01510-z.

[12] B. H. Weinberg *et al.*, "High-performance chemical- and light-inducible recombinases in mammalian cells and mice," *Nature Communications*, vol. 10, 2019, doi: 10.1038/s41467-019-12800-7.

[13] J. C. Tran, C. J. Kuffner, A. C. Reilly, Q. Le, and J. T. Ngo, "ProNotch converts extracellular protease activity into programmable transcriptional outputs," *bioRxiv (preprint)*, 2026, doi: 10.64898/2026.06.01.728985.

[14] M. J. Lajoie *et al.*, "Designed protein logic to target cells with precise combinations of surface antigens," *Science*, vol. 369, 2020, doi: 10.1126/science.aba6527.

[15] M. Bianchi *et al.*, "Logic-Gated Switch-DARPin T Cell Engager with CD2 Co-Stimulation for Improved Safety and Efficacy in MSLN and EpCAM Co-Expressing Ovarian Cancer – MP0632," *Cancer Research*, vol. 86, p. 1624, 2026, doi: 10.1158/1538-7445.am2026-1624.

[16] F. Cattaruzza *et al.*, "Precision-activated T-cell engagers targeting HER2 or EGFR and CD3 mitigate on-target, off-tumor toxicity for immunotherapy in solid tumors," *Nature Cancer*, vol. 4, 2023, doi: 10.1038/s43018-023-00536-9.

[17] B. Salzer *et al.*, "Engineering AvidCARs for combinatorial antigen recognition and reversible control of CAR function," *Nature Communications*, vol. 11, 2020, doi: 10.1038/s41467-020-17970-3.

[18] C. Gu *et al.*, "JMB2403, a potential best-in-class PD-1-dependent IL2Rβγ-activating tri-specific antibody for safe and potent immunotherapy," *mAbs*, vol. 18, 2026, doi: 10.1080/19420862.2026.2709956.

[19] H. B. S. Griffiths *et al.*, "A bispecific antibody–drug conjugate targeting CD7 and CD33 shows anti-tumor activity and improved tumor selectivity in an aggressive subtype of acute myeloid leukemia," *mAbs*, vol. 17, 2025, doi: 10.1080/19420862.2025.2551205.

[20] C. Grandclément *et al.*, "Development of ISB 1442, a CD38 and CD47 bispecific biparatopic antibody innate cell modulator for the treatment of multiple myeloma," *Nature Communications*, vol. 15, 2024, doi: 10.1038/s41467-024-46310-y.

[21] J. Loyau *et al.*, "Biparatopic binding of ISB 1442 to CD38 in trans enables increased cell antibody density and increased avidity," *mAbs*, vol. 17, 2025, doi: 10.1080/19420862.2025.2457471.

[22] E. Pham *et al.*, "AMG 305, a dual targeting BiTE molecule with selective activity for solid tumors that co-express CDH3 and MSLN," *Cancer Research*, vol. 83, p. ND06, 2023, doi: 10.1158/1538-7445.am2023-nd06.

[23] D. M. Dicara *et al.*, "Development of T-cell engagers selective for cells co-expressing two antigens," *mAbs*, vol. 14, 2022, doi: 10.1080/19420862.2022.2115213.

[24] B. J. Yachnin *et al.*, "Massively parallel, computationally guided design of a proenzyme," *Proceedings of the National Academy of Sciences*, vol. 119, 2022, doi: 10.1073/pnas.2116097119.

[25] F. Qi *et al.*, "Hypoxia-activated ADCC-enhanced humanized anti-CD147 antibody for liver cancer imaging and targeted therapy with improved selectivity," *MedComm*, vol. 5, 2024, doi: 10.1002/mco2.512.

[26] M. Kamata-Sakurai *et al.*, "Antibody to CD137 Activated by Extracellular Adenosine Triphosphate Is Tumor Selective and Broadly Effective In Vivo without Systemic Immune Activation," *Cancer Discovery*, vol. 11, 2021, doi: 10.1158/2159-8290.CD-20-0328.

[27] A. Pillai *et al.*, "De novo design of allosterically switchable protein assemblies," *Nature*, vol. 632, 2024, doi: 10.1038/s41586-024-07813-2.

[28] C. Claus *et al.*, "Tumor-targeted 4-1BB agonists for combination with T cell bispecific antibodies as off-the-shelf therapy," *Science Translational Medicine*, vol. 11, 2019, doi: 10.1126/scitranslmed.aav5989.

[29] M. Trüb *et al.*, "Fibroblast activation protein-targeted-4-1BB ligand agonist amplifies effector functions of intratumoral T cells in human cancer," *Journal for ImmunoTherapy of Cancer*, vol. 8, 2020, doi: 10.1136/jitc-2019-000238.

[30] A. Muik *et al.*, "Preclinical Characterization and Phase I Trial Results of a Bispecific Antibody Targeting PD-L1 and 4-1BB (GEN1046) in Patients with Advanced Refractory Solid Tumors," *Cancer Discovery*, vol. 12, 2022, doi: 10.1158/2159-8290.cd-21-1345.

[31] S. Ye *et al.*, "A Bispecific Molecule Targeting CD40 and Tumor Antigen Mesothelin Enhances Tumor-Specific Immunity," *Cancer Immunology Research*, vol. 7, 2019, doi: 10.1158/2326-6066.cir-18-0805.

[32] T. Fink *et al.*, "Design of fast proteolysis-based signaling and logic circuits in mammalian cells," *Nature Chemical Biology*, vol. 15, 2019, doi: 10.1038/s41589-018-0181-6.

[33] B. Zakeri *et al.*, "Peptide tag forming a rapid covalent bond to a protein, through engineering a bacterial adhesin," *Proceedings of the National Academy of Sciences*, vol. 109, 2012, doi: 10.1073/pnas.1115485109.

[34] Z. Chen *et al.*, "De novo design of protein logic gates," *Science*, vol. 368, 2020, doi: 10.1126/science.aay2790.

[35] M. E. Allen *et al.*, "An AND-Gated Drug and Photoactivatable Cre-loxP System for Spatiotemporal Control in Cell-Based Therapeutics," *ACS Synthetic Biology*, vol. 8, 2019, doi: 10.1021/acssynbio.9b00175.

[36] V. Schellenberger *et al.*, "A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner," *Nature Biotechnology*, vol. 27, 2009, doi: 10.1038/nbt.1588.

[37] "Revitope–third-generation T cell engager immunotherapies," *Nature (Biopharma Dealmakers supplement, Advertisement Feature – company-authored, NOT peer-reviewed)*, 2021.

[38] B. Irving, "Abstract ND03: AMX-818: Next generation conditionally active, masked T-cell engager for the treatment of HER2-expressing solid tumors," *Cancer Research*, vol. 82, 2022, doi: 10.1158/1538-7445.am2022-nd03.

[39] J. S. De Bono *et al.*, "Preliminary phase 1 dose escalation results of VIR-5500 (AMX-500), a dual-masked PRO-XTEN T-cell engager, in metastatic castration resistant prostate cancer (mCRPC)." *Journal of Clinical Oncology*, vol. 44, 2026, doi: 10.1200/jco.2026.44.7_suppl.17.

[40] T. R. DiRaimondo, N. Budimir, L. Ma, S. Shenhav, *et al.*, "Preclinical Development of an EGFR-Targeted Tumor-Activated T Cell Engager with Enhanced Safety to Activity Multiple and Pharmacokinetics Profile," *SITC Annual Meeting (poster)*, 2022, Available: https://www.januxrx.com/wp-content/uploads/2022/11/DiRaimondo_EGFR-poster_Final_SITC2022.pdf

[41] G. Liu *et al.*, "Abstract 2869: ADG138, a novel HER2×CD3 POWERbody™ integrating bispecific TCE with precision masking to control cytokine release syndrome and on-target off-tumor toxicity for single agent and combination therapies in HER2-expressing solid tumors," *Cancer Research*, vol. 82, 2022, doi: 10.1158/1538-7445.am2022-2869.

[42] A. Synan *et al.*, "A bispecific antibody-drug conjugate targeting pCAD and CDH17 has antitumor activity and improved tumor-specificity," *mAbs*, vol. 17, 2025, doi: 10.1080/19420862.2024.2441411.

[43] H.-H. Oberg *et al.*, "Tumor-restricted activation of Vγ9Vδ2 T cells via bispecific Evobodies: a novel strategy for safe and potent immunotherapy in ovarian cancer," *Frontiers in Immunology*, vol. 16, 2025, doi: 10.3389/fimmu.2025.1628501.

[44] P. Zhao *et al.*, "Improving dual targeting selectivity in T-cell engagers via synapse-gated and affinity-tuned trispecific antibody design," *mAbs*, vol. 17, 2025, doi: 10.1080/19420862.2025.2570748.

[45] M. Poreba, "Protease-activated prodrugs: strategies, challenges, and future directions," *The FEBS Journal*, vol. 287, 2020, doi: 10.1111/febs.15227.

[46] C. Bleuez, W. F. Koch, C. Urbach, F. Hollfelder, and L. Jermutus, "Exploiting protease activation for therapy," *Drug Discovery Today*, vol. 27, 2022, doi: 10.1016/j.drudis.2022.03.011.

[47] V. Boni *et al.*, "Praluzatamab Ravtansine, a CD166-Targeting Antibody–Drug Conjugate, in Patients with Advanced Solid Tumors: An Open-Label Phase I/II Trial," *Clinical Cancer Research*, vol. 28, 2022, doi: 10.1158/1078-0432.ccr-21-3656.

[48] E. J. Hsu *et al.*, "A cytokine receptor-masked IL2 prodrug selectively activates tumor-infiltrating lymphocytes for potent anti-tumor therapy," *Nature Communications*, vol. 12, 2021, doi: 10.1038/s41467-021-22980-w.

[49] G. Liu, X. She, Y. Li, F. Du, and P. Luo, "Abstract 1853: A novel anti-CTLA-4 checkpoint inhibitor prodrug to address on-target off-tumor toxicity for cancer immunotherapy," *Cancer Research*, vol. 81, 2021, doi: 10.1158/1538-7445.AM2021-1853.

[50] D. Li *et al.*, "578P Increased therapeutic index of muzastotug (ADG126), a masked anti-CTLA-4 antibody, in combination with pembrolizumab (pembro) enables significant clinical benefits and supports further clinical development in patients with metastatic MSS CRC," *Annals of Oncology*, vol. 35, 2024, doi: 10.1016/j.annonc.2024.08.647.

[51] B. Cai *et al.*, "ADG152, a Novel CD20xCD3 T Cell Engager Prodrug with Enhanced Therapeutic Index, Demonstrates Strong Anti-Tumor Activity with Improved Safety," *Blood*, vol. 138, 2021, doi: 10.1182/blood-2021-147573.

[52] G. Liu *et al.*, "Abstract 2868: ADG206, an anti-CD137 agonistic POWERbody with tailor-made efficacy and safety profiles by strong crosslinking and tumor selective activation for single and combinational cancer immunotherapy," *Cancer Research*, vol. 82, 2022, doi: 10.1158/1538-7445.am2022-2868.

[53] A. Panchal *et al.*, "COBRA™: a highly potent conditionally active T cell engager engineered for the treatment of solid tumors," *mAbs*, vol. 12, 2020, doi: 10.1080/19420862.2020.1792130.

[54] D. E. Dettling *et al.*, "Regression of EGFR positive established solid tumors in mice with the conditionally active T cell engager TAK-186," *Journal for ImmunoTherapy of Cancer*, vol. 10, 2022, doi: 10.1136/jitc-2021-004336.

[55] J. Krakow *et al.*, "Structural arrangement of the VH and VL domains in the COBRA™ T-cell engaging single-chain diabody," *Antibody Therapeutics*, vol. 5, 2022, doi: 10.1093/abt/tbab028.

[56] M. Escobar-Rosales *et al.*, "De Novo Design of Peptide Masks Enables Rapid Generation of Conditionally-Active Miniprotein Binders," *Journal of the American Chemical Society*, vol. 147, 2025, doi: 10.1021/jacs.5c16108.

[57] A. E. Vlahos *et al.*, "Protease-controlled secretion and display of intercellular signals," *Nature Communications*, vol. 13, 2022, doi: 10.1038/s41467-022-28623-y.

[58] Y. Liu *et al.*, "A generalizable approach for programming protease-responsive conformationally inhibited artificial transcriptional factors," *Nature Communications*, vol. 16, 2025, doi: 10.1038/s41467-025-59828-6.

[59] S. Xia *et al.*, "Synthetic protein circuits for programmable control of mammalian cell death," *Cell*, vol. 187, 2024, doi: 10.1016/j.cell.2024.03.031.

[60] T. DiRaimondo *et al.*, "Preclinical Activity and Safety Profile of JANX007, a Novel PSMA-Targeting Tumor-Activated T Cell Engager for Treatment of Metastatic Castration-Resistant Prostate Cancer," *SITC Annual Meeting (poster)*, 2022, Available: https://www.januxrx.com/wp-content/uploads/2022/11/DiRaimondo_PSMA-poster_Final_SITC2022.pdf

[61] M. Geiger *et al.*, "Protease-activation using anti-idiotypic masks enables tumor specificity of a folate receptor 1-T cell bispecific antibody," *Nature Communications*, vol. 11, 2020, doi: 10.1038/s41467-020-16838-w.

[62] S. J. Lin *et al.*, "Abstract 933: ProTriTAC is a modular and robust T cell engager prodrug platform with therapeutic index expansion observed across multiple tumor targets," *Cancer Research*, vol. 81, 2021, doi: 10.1158/1538-7445.AM2021-933.

[63] K. S. Kwant *et al.*, "Abstract 2861: TriTAC-XR: An extended-release T cell engager platform designed to minimize cytokine release syndrome," *Cancer Research*, vol. 82, 2022, doi: 10.1158/1538-7445.AM2022-2861.

[64] E. Patel *et al.*, "XTX301, a Tumor-Activated Interleukin-12 Has the Potential to Widen the Therapeutic Index of IL12 Treatment for Solid Tumors as Evidenced by Preclinical Studies," *Molecular Cancer Therapeutics*, vol. 23, 2023, doi: 10.1158/1535-7163.mct-23-0336.

[65] C. J. Nirschl *et al.*, "Discovery of a Conditionally Activated IL-2 that Promotes Antitumor Immunity and Induces Tumor Regression," *Cancer Immunology Research*, vol. 10, 2022, doi: 10.1158/2326-6066.cir-21-0831.

[66] C. J. Nirschl *et al.*, "mWTX-330, an IL-12 INDUKINE Molecule, Activates and Reshapes Tumor-Infiltrating CD8+ T and NK Cells to Generate Antitumor Immunity," *Cancer Immunology Research*, vol. 11, 2023, doi: 10.1158/2326-6066.cir-22-0705.

[67] N. B. Woodall *et al.*, "De novo design of tyrosine and serine kinase-driven protein switches," *Nature Structural & Molecular Biology*, vol. 28, 2021, doi: 10.1038/s41594-021-00649-8.

[68] Q. Cao and J. E. Toettcher, "Design and optimization of a kinase-controlled allosteric switch," *Nature Methods*, vol. 23, 2026, doi: 10.1038/s41592-026-03163-1.

[69] J. Khowsathit, A. Bazzoli, H. Cheng, and J. Karanicolas, "Computational Design of an Allosteric Antibody Switch by Deletion and Rescue of a Complex Structural Constellation," *ACS Central Science*, vol. 6, 2020, doi: 10.1021/acscentsci.9b01065.

[70] K. Deckert, S. J. Budiardjo, L. C. Brunner, S. Lovell, and J. Karanicolas, "Designing Allosteric Control into Enzymes by Chemical Rescue of Structure," *Journal of the American Chemical Society*, vol. 134, 2012, doi: 10.1021/ja301409g.

[71] S. J. Budiardjo *et al.*, "Full and Partial Agonism of a Designed Enzyme Switch," *ACS Synthetic Biology*, vol. 5, 2016, doi: 10.1021/acssynbio.6b00097.

[72] T. Plaper *et al.*, "Designed allosteric protein logic," *Cell Discovery*, vol. 10, 2024, doi: 10.1038/s41421-023-00635-y.

[73] S.-J. Kellmann, S. Dübel, and H. Thie, "A strategy to identify linker-based modules for the allosteric regulation of antibody-antigen binding affinities of different scFvs," *mAbs*, vol. 9, 2017, doi: 10.1080/19420862.2016.1277302.

[74] T. Nemoto, H. Kuroi, M. Shida-Kawazoe, and T. Tachibana, "Quantitative Investigation of ATP Switch Concept of STA551 Through Combined In Vivo Tissue Distribution Studies and Physiologically Based Pharmacokinetic Modeling," *The AAPS Journal*, vol. 28, 2026, doi: 10.1208/s12248-026-01238-7.

[75] H. Hayashi *et al.*, "ROSE12, a novel anti-CTLA-4 FcγRs binding-enhanced antibody activated by extracellular adenosine triphosphate, shows tumor-selective regulatory T-cell depletion and antitumor efficacy without systemic immune activation," *Journal for ImmunoTherapy of Cancer*, vol. 14, 2026, doi: 10.1136/jitc-2025-013397.

[76] S. N. Ho, S. R. Biggar, D. M. Spencer, S. L. Schreiber, and G. R. Crabtree, "Dimeric ligands define a role for transcriptional activation domains in reinitiation," *Nature*, vol. 382, 1996, doi: 10.1038/382822a0.

[77] A. Di Stasi *et al.*, "Inducible Apoptosis as a Safety Switch for Adoptive Cell Therapy," *New England Journal of Medicine*, vol. 365, 2011, doi: 10.1056/nejmoa1106152.

[78] A. J. Martinko *et al.*, "Switchable assembly and function of antibody complexes in vivo using a small molecule," *Proceedings of the National Academy of Sciences*, vol. 119, 2022, doi: 10.1073/pnas.2117402119.

[79] E. Rihtar *et al.*, "Ligand-induced assembly of antibody variable fragments for the chemical regulation of biological processes," *Cell Chemical Biology*, vol. 32, 2025, doi: 10.1016/j.chembiol.2025.01.007.

[80] I. Melero *et al.*, "A first-in-human study of the fibroblast activation protein–targeted, 4-1BB agonist RO7122290 in patients with advanced solid tumors," *Science Translational Medicine*, vol. 15, 2023, doi: 10.1126/scitranslmed.abp9229.

[81] A. F. Labrijn *et al.*, "Efficient generation of stable bispecific IgG1 by controlled Fab-arm exchange," *Proceedings of the National Academy of Sciences*, vol. 110, 2013, doi: 10.1073/pnas.1220145110.

[82] U. Jung *et al.*, "892 ABL503 (TJ-L14B), PD-L1x4–1BB bispecific antibody induces superior anti-tumor activity by PD-L1-dependent 4–1BB activation with the increase of 4–1BB+CD8+ T cells in tumor microenvironment," *Regular and Young Investigator Award Abstracts*, 2021, doi: 10.1136/jitc-2021-sitc2021.892.

[83] S. H. Jeon *et al.*, "Anti-4-1BB×PDL1 Bispecific Antibody Reinvigorates Tumor-Specific Exhausted CD8+ T Cells and Enhances the Efficacy of Anti-PD1 Blockade," *Clinical Cancer Research*, vol. 30, 2024, doi: 10.1158/1078-0432.ccr-23-2864.

[84] H. K. Binz *et al.*, "High-affinity binders selected from designed ankyrin repeat protein libraries," *Nature Biotechnology*, vol. 22, 2004, doi: 10.1038/nbt962.

[85] I. Saltarella *et al.*, "Improvement of daratumumab- or elotuzumab-mediated NK cell activity by the bi-specific 4-1BB agonist, DARPin -FAPx4–1BB: A preclinical study in multiple myeloma," *Biomedicine & Pharmacotherapy*, vol. 176, 2024, doi: 10.1016/j.biopha.2024.116877.

[86] N. Steeghs *et al.*, "Tumor-localized CD40 agonism with MP0317, a FAP x CD40 DARPin, reprograms the tumor microenvironment in patients with advanced solid tumors: an open-label, nonrandomized, dose-escalation phase 1 study," *Nature Cancer*, vol. 7, 2026, doi: 10.1038/s43018-026-01150-1.

[87] C. Rothe and A. Skerra, "Anticalin® Proteins as Therapeutic Agents in Human Diseases," *BioDrugs*, vol. 32, 2018, doi: 10.1007/s40259-018-0278-1.

[88] M. J. Hinner *et al.*, "Tumor-Localized Costimulatory T-Cell Engagement by the 4-1BB/HER2 Bispecific Antibody-Anticalin Fusion PRS-343," *Clinical Cancer Research*, vol. 25, 2019, doi: 10.1158/1078-0432.ccr-18-3654.

[89] G. Wozniak-Knopp *et al.*, "Introducing antigen-binding sites in structural loops of immunoglobulin constant domains: Fc fragments with engineered HER2/neu-binding sites and antibody properties," *Protein Engineering, Design and Selection*, vol. 23, 2010, doi: 10.1093/protein/gzq005.

[90] M. A. Lakins *et al.*, "FS222, a CD137/PD-L1 Tetravalent Bispecific Antibody, Exhibits Low Toxicity and Antitumor Activity in Colorectal Cancer Models," *Clinical Cancer Research*, vol. 26, 2020, doi: 10.1158/1078-0432.ccr-19-2958.

[91] C. Geuijen *et al.*, "A human CD137×PD-L1 bispecific antibody promotes anti-tumor immunity via context-dependent T cell costimulation and checkpoint blockade," *Nature Communications*, vol. 12, 2021, doi: 10.1038/s41467-021-24767-5.

[92] R. A. Hernandez-Lopez *et al.*, "T cell circuits that sense antigen density with an ultrasensitive threshold," *Science*, vol. 371, 2021, doi: 10.1126/science.abc1855.

[93] J. H. Cho, A. Okuma, K. Sofjan, S. Lee, J. J. Collins, and W. W. Wong, "Engineering advanced logic and distributed computing in human CAR immune cells," *Nature Communications*, vol. 12, 2021, doi: 10.1038/s41467-021-21078-7.

[94] M. Y. Siddiqui *et al.*, "Tunable universal OR-gated CAR T cells for AML," *iScience*, vol. 29, 2026, doi: 10.1016/j.isci.2026.117008.

[95] H. Wajant, "Principles of antibody-mediated TNF receptor activation," *Cell Death & Differentiation*, vol. 22, 2015, doi: 10.1038/cdd.2015.109.

[96] N. Duru *et al.*, "Selective targeting of metastatic ovarian cancer using an engineered anthrax prodrug activated by membrane-anchored serine proteases," *Proceedings of the National Academy of Sciences*, vol. 119, 2022, doi: 10.1073/pnas.2201423119.

[97] T. Bridge, S. A. Shaikh, P. Thomas, J. Botta, P. J. McCormick, and A. Sachdeva, "Site-Specific Encoding of Photoactivity in Antibodies Enables Light-Mediated Antibody–Antigen Binding on Live Cells," *Angewandte Chemie International Edition*, vol. 58, 2019, doi: 10.1002/anie.201908655.

[98] Y. Zhao *et al.*, "Switchable immune modulator for tumor-specific activation of anticancer immunity," *Science Advances*, vol. 7, 2021, doi: 10.1126/sciadv.abg7291.

[99] T. W. McKeithan, "Kinetic proofreading in T-cell receptor signal transduction." *Proceedings of the National Academy of Sciences*, vol. 92, 1995, doi: 10.1073/pnas.92.11.5042.

[100] R. A. Langan *et al.*, "De novo design of bioactive protein switches," *Nature*, vol. 572, 2019, doi: 10.1038/s41586-019-1432-8.

[101] Y. Xia *et al.*, "The Designability of Protein Switches by Chemical Rescue of Structure: Mechanisms of Inactivation and Reactivation," *Journal of the American Chemical Society*, vol. 135, 2013, doi: 10.1021/ja407644b.

[102] E. H. Kellogg, A. Leaver-Fay, and D. Baker, "Role of conformational sampling in computing mutation-induced changes in protein structure and stability," *Proteins: Structure, Function, and Bioinformatics*, vol. 79, 2010, doi: 10.1002/prot.22921.

[103] J. Schymkowitz, J. Borg, F. Stricher, R. Nys, F. Rousseau, and L. Serrano, "The FoldX web server: an online force field," *Nucleic Acids Research*, vol. 33, 2005, doi: 10.1093/nar/gki387.

[104] J. Dauparas *et al.*, "Robust deep learning–based protein sequence design using ProteinMPNN," *Science*, vol. 378, 2022, doi: 10.1126/science.add2187.

[105] J. Dauparas *et al.*, "Atomic context-conditioned protein sequence design using LigandMPNN," *Nature Methods*, vol. 22, 2025, doi: 10.1038/s41592-025-02626-1.

[106] Z. Lin *et al.*, "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science*, vol. 379, 2023, doi: 10.1126/science.ade2574.

[107] X. Wu *et al.*, "Multivalency Enables Signal Processing at Single Protein Level," *Journal of the American Chemical Society*, vol. 147, 2025, doi: 10.1021/jacs.5c11096.

[108] J. L. Watson *et al.*, "De novo design of protein structure and function with RFdiffusion," *Nature*, vol. 620, 2023, doi: 10.1038/s41586-023-06415-8.

[109] M. Pacesa *et al.*, "One-shot design of functional protein binders with BindCraft," *Nature*, vol. 646, 2025, doi: 10.1038/s41586-025-09429-6.

[110] H. Stark *et al.*, "BoltzGen: Toward Universal Binder Design," 2025, doi: 10.1101/2025.11.20.689494.

[111] Escalante Bio, "Mosaic: a JAX framework for multi-objective, gradient-based protein binder design." Software. Available: https://github.com/escalante-bio/mosaic

[112] J. Jumper *et al.*, "Highly accurate protein structure prediction with AlphaFold," *Nature*, vol. 596, 2021, doi: 10.1038/s41586-021-03819-2.

[113] J. Abramson *et al.*, "Accurate structure prediction of biomolecular interactions with AlphaFold 3," *Nature*, vol. 630, 2024, doi: 10.1038/s41586-024-07487-w.

[114] D. del Alamo, D. Sala, H. S. Mchaourab, and J. Meiler, "Sampling alternative conformational states of transporters and receptors with AlphaFold2," *eLife*, vol. 11, 2022, doi: 10.7554/eLife.75751.

[115] X. Wang, P. Lupardus, S. L. LaPorte, and K. C. Garcia, "Structural Biology of Shared Cytokine Receptors," *Annual Review of Immunology*, vol. 27, 2009, doi: 10.1146/annurev.immunol.24.021605.090616.